Adapting biological mechanisms for agentic protein design

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Abstract

In silico protein design models address only one step toward achieving a desired biological function. Connecting binder design with target selection, binding-site analysis, and mechanism-specific assessment remains a fragmented workflow that requires expert input. We present BindClaw, an agent-assisted workflow that, starting from a user supplied prompt, streamlines functional protein design, including target discovery, multi-generator design, computational assay construction and iterative refinement. Given a starting prompt or paper, BindClaw can operate autonomously using agents to construct a mechanism-specific computational assay. It can then iteratively leverage protein design backends to discover functional candidates tested through computational assays. In silico campaigns illustrate how agents revise target interfaces, adjust protein design backend settings, and revisit overall design specifications in response to computational results. In a human designer-assisted campaign, 15 of 18 designs bound calcium-loaded calmodulin in the wetlab, with biolayer interferometry (BLI) replicate medians in the subnanomolar and low-nanomolar range. Under the tested conditions, 13 of the 15 binders showed no detectable binding to non-calcium-loaded calmodulin, while the remaining two bound more strongly to calcium-loaded calmodulin. BindClaw distills biological objectives into testable computational assays, leveraging protein design backends to obtain binders that respond to functional constraints, in turn reducing the friction from biological hypothesis to testable binder.

Keywords: *Protein design, Functional binders, Agentic AI, Mechanism-guided design, Computational assays*

Project page: https://research.nvidia.com/labs/dbr/assets/pages/bindclaw

1. Introduction

Protein interactions govern how cells transmit signals, assemble molecular complexes and respond to changing conditions (Pawson & Nash, 2003; Alberts, 1998; Changeux & Edelstein, 2005). Designed binders offer a way to intervene in these processes by occupying a native interaction site (Pacesa et al., 2025), bringing molecular partners together (Huang et al., 2025; Silva et al., 2019) or preferentially recognizing a ligand-bound target surface (Marchand et al., 2025). Achieving these effects requires more than a structurally plausible interface: the binder must engage the appropriate molecular context in a way that supports the intended mechanism.

Advances in generative modeling, sequence design and structure prediction have expanded the ability to construct candidate proteins and assess their structural compatibility (Watson et al., 2023; Dauparas et al., 2022; Jumper et al., 2021; Pacesa et al., 2025). Candidate-design methods include BindCraft (Pacesa et al., 2025), BoltzDesign1 (Cho et al., 2025a), PXDesign (Team et al., 2025), BoltzGen (Stark et al., 2025), Proteina-Complexa (Didi et al., 2026) and RFD4-Proteina (Corley et al., 2026). However, confident in silico predictions of molecular complexes do not establish whether a

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binder competes with a native partner, distinguishes alternative target states or supports a productive molecular assembly. Each of these design objectives requires a different comparison, and establishing the resulting biological response requires functional experiments. Connecting molecular recognition to biological function therefore calls for tailored assessments (Huettemann & Kortemme, 2026). For example, Park et al. (2025) used an AlphaFold 3 competition test to help select a binder targeting the MLH1–PMS2 interface, then experimentally demonstrated mismatch-repair suppression and improved prime editing. This provides a system-specific precedent for using computational competition tests to guide selection of functional inhibitors.

Constructing these assessments involves choices throughout the design workflow: researchers must identify relevant constructs, partners and states, map biological evidence onto candidate binding sites, and select measurements and controls that can test the proposed mechanism. The resulting evidence must also inform the next design decision. An unsuccessful candidate may motivate a different binding site, generation setting or sequence (Bennett et al., 2023), whereas an inadequate assay requires revision of the comparison itself. Distinguishing these changes is essential: a higher score after revising an assay does not, by itself, show that the design improved.

Agent-assisted research offers a means of coordinating these decisions. For instance, Paper2Agent and the Virtual Lab connect published methods or expert guidance to executable research tasks, while T-REX, AutoBinder and Agent Rosetta organize aspects of iterative protein design (Miao et al., 2025; Swanson et al., 2025; Jeon et al., 2026; Ge et al., 2026; Teneggi et al., 2026). A recent study from muni.bio adapted the Autoresearch approach introduced by Andrej Karpathy (Karpathy, 2026) and varied Proteina-Complexa (Didi et al., 2026) settings to improve ipSAE scores (Dunbrack, 2025) from Boltz-2 predictions (Muni, 2026). Its subsequent experimental report describes nine measurable TREM2 binders among ten submitted designs, including three with subnanomolar replicate-mean affinities (muni, 2026).

Here, we present BindClaw, an agentic workflow to connect target discovery and computational assay construction with iterative binder design (Fig. 1). Through our studies, we show that agentic approaches can leverage expert computational assays, and derive new ones, to mimic functional assays from expert literature. We validate functional design through a human-guided wet-lab campaign, showing preferential binding under calcium-loaded versus calcium-depleted assay conditions. Furthermore, we show that agents can adjust for generation backend hotspot preferences, and reason through historical results to fine-tune specification and generation settings. Finally, BindClaw can be run end-to-end to connect biological objectives, generation backends, and mechanism-specific assessment in an end-to-end design workflow, guiding binder generation toward the requirements of their intended biological function.

2. Results

BindClaw is an agentic workflow for functional binder design (Fig. 1A). From a user prompt (“Design objective”) and optional paper, it autonomously derives a specification (“Specification”), generates candidates (“Design”) and performs computational assessments (“Assessment”).

Specification defines the biological objective, molecular inputs and comparisons needed to assess the intended function. The *intake* stage extracts the target, desired function, specificity requirements and design constraints. In the programmed workflow, intake uses rule-based parsing with optional LLM refinement. *Target ID* retains a named target or uses literature and database searches to identify one, retrieves UniProt annotations, and selects a structure from RCSB PDB or AlphaFold DB (Consortium, 2015; Burley et al., 2019; Varadi et al., 2024). Input preparation checks chains, sequence coverage and biological assembly. If a suitable natural-chain template is unavailable, prediction tools

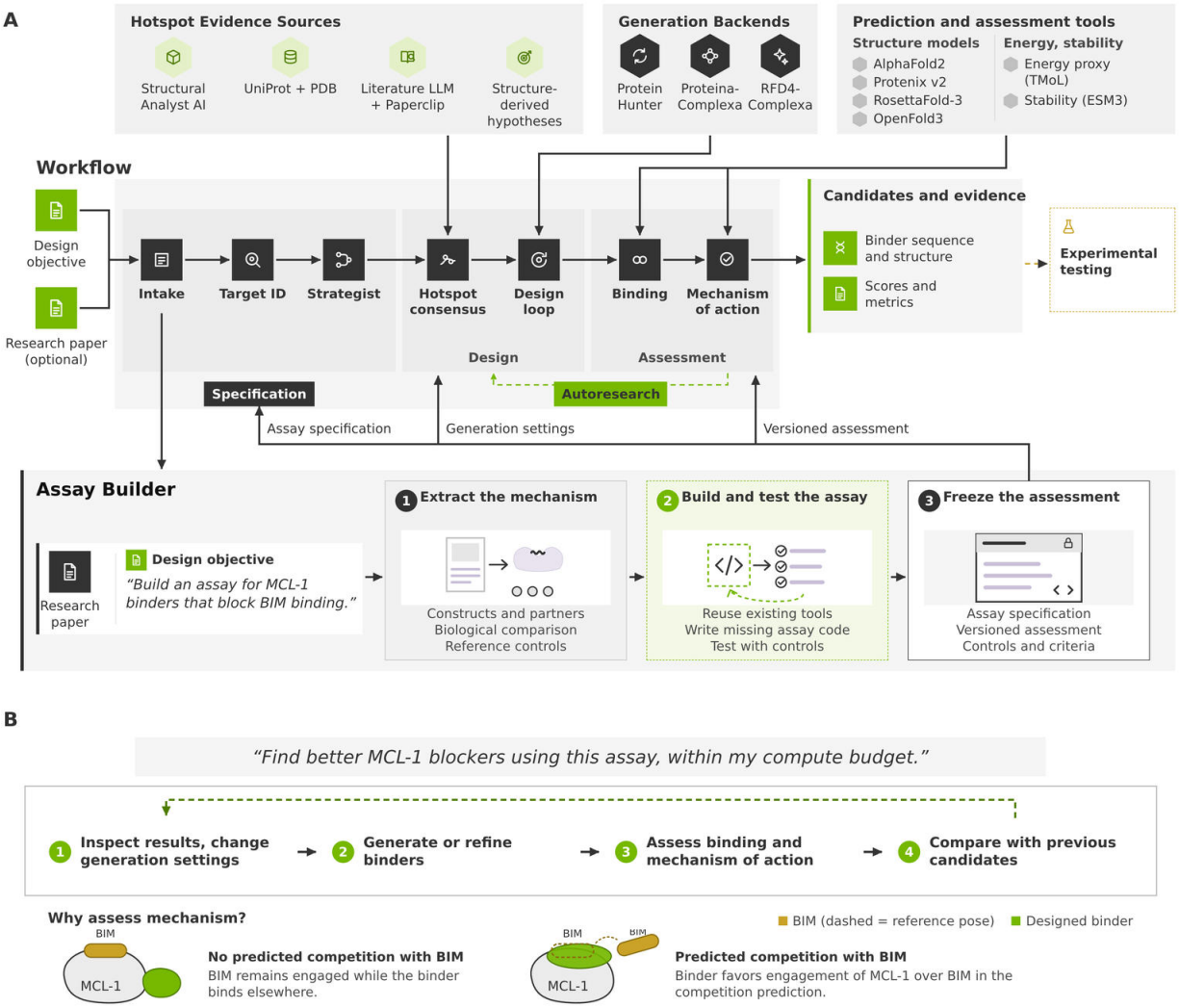


Figure 1 | BindClaw connects a biological objective to protein design and assessment. (A) The assay builder uses the target, desired mechanism and supplied evidence to build a computational assay. (B) Autoresearch generates and revises candidates against that fixed assay. The MCL-1 example shows how the assay checks both target binding and occupation of BIM's binding site. Candidate sequences and assessment results guide experimental follow-up.

can generate one using a multiple sequence alignment (MSA) of homologous sequences. The *strategist* agent uses the objective and target evidence to propose the biological mechanism and relevant native partners, alternative states or molecular arrangements to compare during *Assessment*.

The *Assay Builder* translates these requirements into a computationally executable assay, specifying how to calculate measurements and apply assessment criteria. It selects existing mechanism-specific measurements or uses a coding agent to implement and test missing measurements before freezing the assay. For paper-derived objectives, text and assets extracted by Paper2Agent (Miao et al., 2025) and literature retrieved through PaperClip (GXL, 2025) provide evidence for the molecular inputs, controls and comparisons (App. A.2). Thus, the specification defines what the design should achieve, while the Assay Builder implements how candidates will be assessed.

Agent roles use LLM calls and tools to interpret evidence and propose choices. Software services validate inputs, execute model calls, calculate measurements and apply the recorded criteria. Stages exchange structured records and workspace files containing specifications, target evidence, plans and assay definitions. Downstream agents receive the objective and relevant saved outputs as context.

Within *Design*, the *Hotspot consensus* agent reasons through hotspot residue proposals from experimental structures or predicted models, UniProt/PDB annotations and literature, grouping them into spatially coherent binding sets (App. B). These sets provide alternative hypotheses about where binding could support the intended mechanism. The *Design loop* agent takes the targets, hotspots and other generation constraints into the input format required by each generation backend, including RFD4-Proteina, Proteina-Complexa and Protein Hunter (Corley et al., 2026; Didi et al., 2026; Cho et al., 2025b).

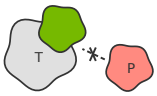
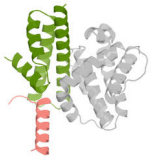
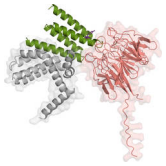
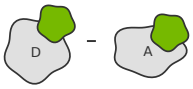
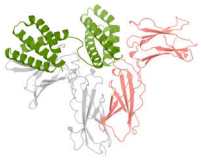
Within *Assessment*, the *Binding* agent evaluates confidence and agreement between generated designs, and separately co-folded poses, while the *Mechanism of action* applies the comparisons defined in the assay (App. A.1).

The *Autoresearch* loop (Fig. 1B) involves an agent inspecting binding and mechanism assessments, changing generation settings to produce new binders or revising existing designs through sequence redesign or targeted mutations, and comparing the resulting assessments with those of earlier designs. Execution software runs the requested generation and assessment jobs and collects their results, recording each design's sequence, any earlier design from which it was derived, model settings, GPU usage and execution time. Based on the results, the agent adjusts generation settings and records the changes alongside the resulting designs and assessments (campaign criteria in App. E). Designs that pass the full computational screen are then collected and reported.

2.1. Expert predefined assays assess distinct biological objectives

We predefined computational assays mimicking biological modes of action for inhibition, protein recruitment, stabilization, state-selective binding and receptor agonism (Fig. 2A). These assays were guided by scientific literature in the choice of molecules and comparisons used. For instance, the inhibitor assay assesses whether a candidate can occupy the native BIM-binding site on MCL-1 with BIM present (Fire et al., 2010). This follows an interface-blocking strategy also used to design an MLH1 binder at the MLH1–PMS2 interface (Park et al., 2025). The degrader assay tests whether a design can engage both BCL-xL and the E3 ligase substrate receptor KLHL20 (Mylemans et al., 2026). The conformational stabilization assay tests whether a binder helps preserve a desired target conformation, illustrated here with a short helical TDP-43 segment (Sun et al., 2025). The state-selective assay compares calcium-free (apo) and calcium-loaded (holo) CaM using differential scoring (Cao et al., 2026). Finally, the receptor agonism assay checks engagement and arrangement of two IL-2 receptor subunits, following the cytokine receptor-geometry design rationale of Expòsit et al. (2025).

A Biological objectives and computational gates

Objective	MoA example	Assessed designs	Computational mechanism checks
Inhibitor / neutralizer MCL-1 / BIM Block BIM binding	 Block partner site Fire et al., 2010		• Engage MCL-1 • Compare binder with BIM • Check BIM-site occupancy
Degrader BCL-xL / KLHL20 Recruit E3 to target	 Recruit target + E3 Mylemans et al., 2026		• Engage target and E3 • Check separate contact regions • Check internal motif placement • Avoid clashes
Stabilizer TDP-43 peptide Stabilize target fold	 Fold rescue Sun et al., 2025		• Contact the target region • Check the binder fold • Assess predicted fold rescue
Allosteric Calmodulin states Favor holo CaM	 Desired vs alternative Cao et al., 2026		• Compare holo and apo states • Check target-state conformations • Score state preference
Agonist IL-2 receptor fusion Activate IL-2 receptors	 Receptor arrangement Expòsit et al., 2025		• Engage both receptors • Check binder-arm assignment • Check receptor arrangement

B MCL-1: from binding-site evidence to a designed complex

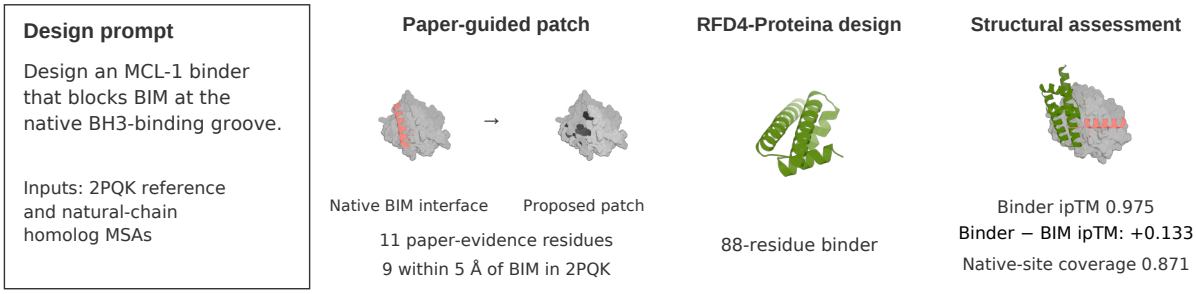


Figure 2 | Biological objectives and computational gates. (A) Mechanism-specific assays and representative designs: target silver, binder green, partner coral. The degrader example is the F63Y design (target/E3 ipTM 0.881/0.601, footprint overlap 0.053). The stabilizer row shows TDP-43 fold rescue. The IL-2 fusion failed engagement. (B) MCL-1/BIM workflow: nine of 11 proposed residues lie within 5 Å of BIM in 2PQK. The dark-gray patch and confidence comparisons assess site compatibility, not displacement. The agonist design follows receptor-arrangement principles (Expòsit et al., 2025; Wang et al., 2005).

BindClaw reuses generation, prediction and measurement tools from the predefined assays when constructing new campaigns from a user's design objective or input paper. It can add additional evaluation functions when a new mechanism requires a comparison outside the predefined set.

A MCL-1/BIM campaign applied the predefined inhibition assay, using MCL-1 as the target, BIM as its native partner and the generated binder as the candidate (Fig. 2B). The functional objective of this campaign is to block the interaction between MCL-1 and its natural partner BIM at the BH3-binding groove (Fire et al., 2010). An 11-residue binding-site hypothesis guided an 88-residue design, which was assessed with both MCL-1 and BIM present (Fig. 2B). The assay evaluated whether the predicted binder occupied the native BIM-binding site with BIM present.

2.2. In silico success rates depend on binding site and generator

BindClaw's workflow allowed us to study whether binding sites with high in silico success rates for one generation backend also achieved high success rates with other backends, by selecting binding site sets and testing different backends autonomously (Fig. 3). Through specification and hotspot consensus, the BindClaw workflow raised six binding-site sets across nine targets, against which designs were generated with four backends: Proteina-Complexa (Didi et al., 2026), RFD4-Proteina (Corley et al., 2026), Protein Hunter (Cho et al., 2025b) and PXDesign (Team et al., 2025). For this benchmark, we additionally included PXDesign as a comparator, integrating its generated candidates into the same assessment pipeline. Standard and beam-search configurations of Proteina-Complexa and RFD4-Proteina, together with one configuration each of Protein Hunter and PXDesign, gave six configurations in total. Protein Hunter used Boltz-2 to guide iterative binder generation and sequence redesign (Cho et al., 2025b; Passaro et al., 2025). Candidates from all design backends were evaluated with the same Protenix-v2 structure prediction protocol (Zhang et al., 2026). For this study, we manually defined the in silico success rate as the fraction of valid evaluations with both ipTM > 0.80 and binder pLDDT > 80. We obtained 6,480 design candidates and evaluations (App. B, Fig. 8).

For PD-L1, expanding set 2 to set 3 added three hotspot residues to a shared ten-residue patch (Fig. 3A). Adding just these three hotspots increased the in silico success rate of Proteina-Complexa beam from 5% to 60% (1 to 12 passes out of 20). RFD4-Proteina beam remained at 100%, while Protein Hunter decreased from 75% to 60% (15 to 12 passes out of 20). PXDesign remained at 45% (9 passes out of 20 at each site).

Targeting EGFR, the in silico success rate of Proteina-Complexa beam was 100% at set 4 and 65% at set 6, whereas RFD4-Proteina beam achieved 10% and 90%, respectively (Fig. 3B). The preferred site therefore reversed between design backends. Protein Hunter achieved 20% and 40% at these EGFR sites, respectively. PXDesign remained at 5% at both sites (20 evaluations per site for each backend).

These examples show that binding-site, hotspot, and generator choices interact: a site that performs well with one generator may perform poorly with another. BindClaw evaluates and automates these searches rather than fixing one site and one generator, allowing it to steer towards increased in silico success.

2.3. Translating published biological objectives into computational tests and designs

We examined whether BindClaw's *Assay Builder* could use a research paper to guide candidate design and computational assessment (Fig. 4). The *Assay Builder* agent uses document-conversion tools from Paper2Agent (Miao et al., 2025) to extract and package a supplied paper's evidence for agents to interpret. Starting from this evidence, an agent then identifies the biological objective and defines

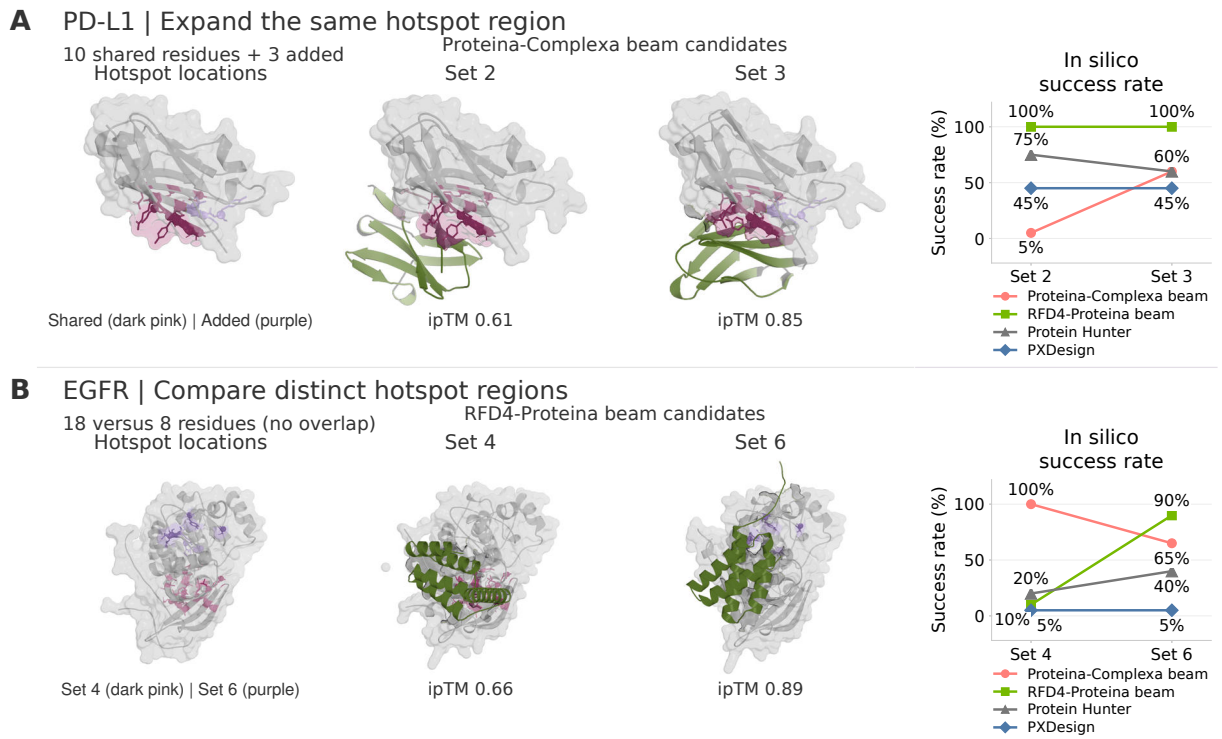


Figure 3 | Binding-site choice affects generators differently. (A) PD-L1 set 3 adds residues 116–118 to set 2's ten hotspots. (B) EGFR sets 4 and 6 contain 18 and eight nonoverlapping hotspots. Generator preferences reverse. Plots include Proteina-Complexa beam, RFD4-Proteina beam, Protein Hunter and PXDesign. Shared PD-L1/set-4 EGFR hotspots are dark pink, added PD-L1/set-6 EGFR hotspots lavender, and binders green. Success requires ipTM > 0.80 and binder pLDDT > 80 (20 valid evaluations per condition). Full 6,480-evaluation benchmark: Fig. 8.

computational comparisons that test features of the mechanism of interest. For example, the Hsp70 activator study combined Rosetta design with AlphaFold2-based screening (Zhang et al., 2025). Its biological objective motivated the DnaK state comparison.

For the IL-2/IL-15 mimic, the assay tested engagement of both receptor subunits and their arrangement (Fig. 4A) (Silva et al., 2019). For the DnaK binder, the agent evaluated the same candidate with ATP- and ADP-state templates (Zhang et al., 2025). The illustrated design scored higher with ATP-state conditioning than with ADP-state conditioning, with interface ipTM values of 0.923 and 0.376 (Fig. 4B). For dual-receptor engagement, the agent joined two binders into one fusion protein to engage both receptor domains (Fig. 4C) (Huang et al., 2025). These two examples show how the agent made the mechanism test more discriminating.

Two small-molecule papers guided assays for DFHBI recognition (Fig. 4D), and digoxigenin selectivity (Fig. 4E). For DFHBI, the agent identified the paper's requirement for a planar Z fluorogen conformation, and built an assay that rejects predicted ligand poses that fail checks for planarity or the required Z configuration. It redesigned the published mFAP1 scaffold (Dou et al., 2018) and screened the candidates with this assay. A separate de novo DFHBI-binding design passed its binding and pose screen with a helical scaffold and preserved the ligand's Z configuration (Fig. 4D). Following Tinberg et al. (2013), the agent generated candidates that passed the computational selectivity screen for digoxigenin over digitoxigenin, progesterone and β-estradiol (Fig. 4E).i

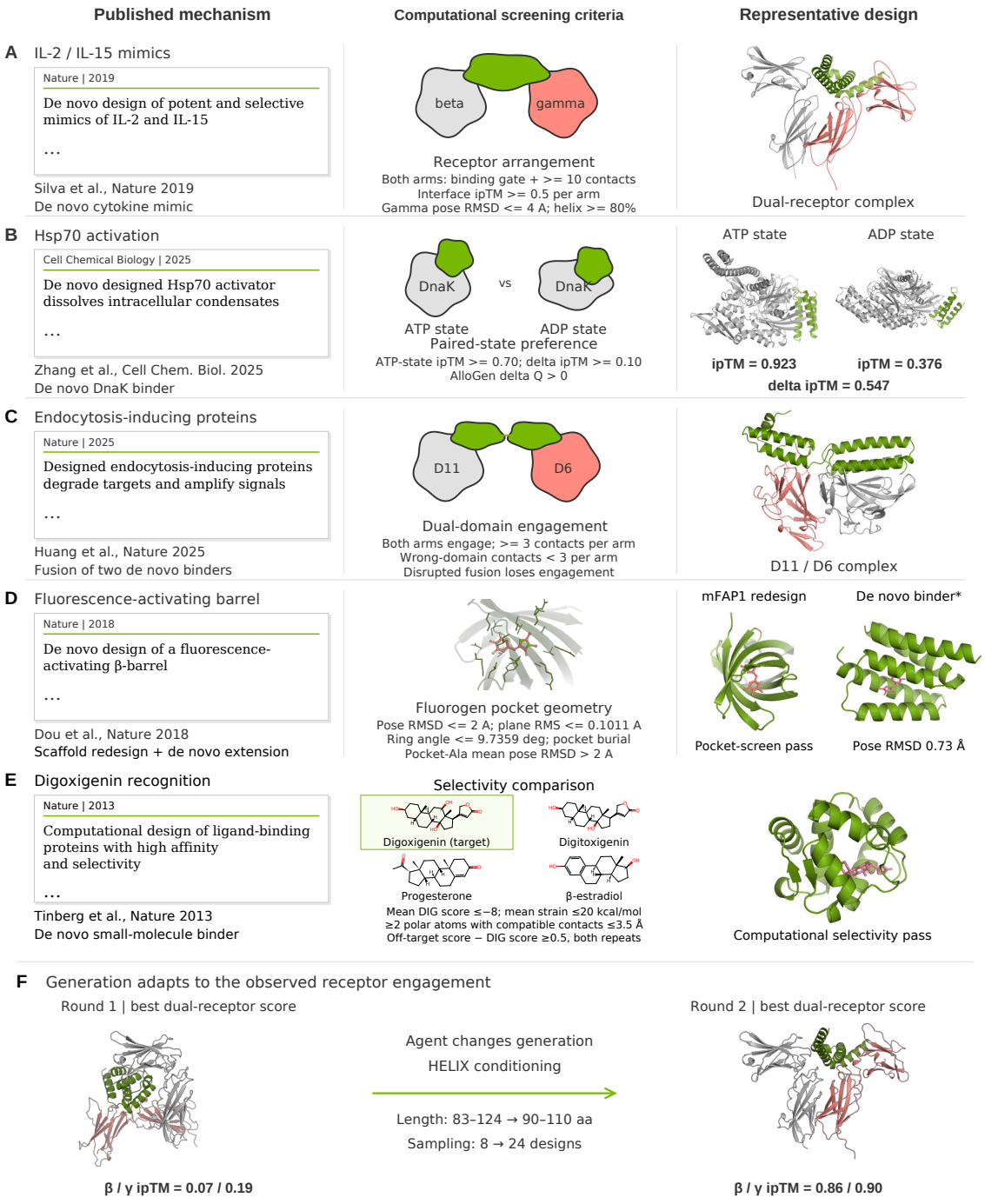


Figure 4 | From published mechanisms to computational hypotheses. Papers, assays and structures show (A) IL-2/IL-15 mimic design, (B) DnaK state comparison, (C) dual-binder fusion with cross-binding and arm-disruption tests, and (D) mFAP1/DFHBI redesign alongside a separate de novo DFHBI binder. The listed pocket criteria apply to the redesign. The de novo example (*) passed its binding/pose screen and retained Z stereochemistry. (E) Binder passes the computational selectivity screen against three off-target ligands. (F) Generation settings revised after weak dual-receptor engagement. Targets are silver, designs green and partners coral.

For IL-2/IL-15, after weak dual-receptor predictions in the first batch, the agent added helical conditioning, narrowed the length range and increased sampling (Fig. 4F). The next batch contained predominantly helical designs, including a candidate with receptor-interface ipTM values of 0.86 and 0.90.

Together, these cases show how the agents within the assay builder translate a biological objective into generation choices and mechanism-specific computational tests (Table 7).

2.4. Autoresearch to revise designs and direct further sampling

BindClaw *Autoresearch* agents run experiments, use the results to form hypotheses, and choose what to test next within a fixed assay and resource budget. For instance, we can prompt a coding agent with: "Given this budget, design binders for <target> and obtain 10 designs that pass the computational screen." The agents can then revise interfaces, expand sampling, keep productive recipes, and return to earlier settings after unsuccessful trials in an autonomous research loop (*autoresearch*) (Karpathy, 2026; Muni, 2026) (Fig. 5).

For BCL-xL/KLHL20, an agent with a protein hunter in the loop generated target binders and, to expose the recruitment motif while preserving target contacts, grafted the KLHL20-recruiting peptide AGLPDLVGA into an internal loop (Mylemans et al., 2026; Chen et al., 2019) (Fig. 5A). It then tested interface substitutions to reduce interference between target and E3 binding, followed by motif extensions and surface variants. The agent combined mutations from variants that had passed individually, then reassessed each combined design. The longer-motif comparison produced no passes, which prompted the agent to continue testing shorter-motif variants. After a second scaffold failed the E3-engagement criterion, it tested 18 motif-flanking substitutions. The F63Y variant reached E3 ipTM 0.601, compared with 0.416 for its parent, and passed the full computational screen. Together, these branches yielded 11 unique passes across two scaffold families under the same assessment criteria.

For SAM/SAH, the agent varied binder lengths to explore different scaffold sizes, and used additional seeds to sample alternative designs at each length. It expanded this initial search before evaluation results were available. Each candidate was independently predicted with both ligands using Protenix v2 (Zhang et al., 2026), allowing the screen to distinguish SAM engagement from a preference for the closely related SAH (Fig. 5B). The illustrated failed design had higher minimum interface PAE with SAM than SAH (3.79 versus 2.13 Å). The passing design showed computational selectivity for SAM over the off-target SAH (minimum interface PAE 1.27 versus 1.75 Å) and met the remaining screen criteria.

For de novo DFHBI design, the agent varied RFD4-Proteina beam-search parameters by increasing sampling from four to eight branches, with 60 steps and beam width one, resulting in three pose failures and one invalid prediction. The agent then returned to four branches with a new seed selection to test whether sampling from a different seed could recover designability, i.e. the agreement between the generated and refolded poses. This produced designs that passed both the designability check and the remaining binding-screen criteria (Fig. 5C).

The agent also reconsidered its choices when additional results weakened its prior hypothesis. For CaM, after several pose failures, the agent proposed using four hotspots instead of six. When more designs passed with six hotspots, it kept six and did not test the four-hotspot proposal (Fig. 5D). For MCL-1/BIM, the competition screen (Fire et al., 2010) already produced full passes in the initial batches. The agent therefore retained the recipe and changed the seed to sample more candidates (Fig. 5E). These campaigns document how results informed revision, continued sampling and deferred hypotheses under fixed assessment criteria.

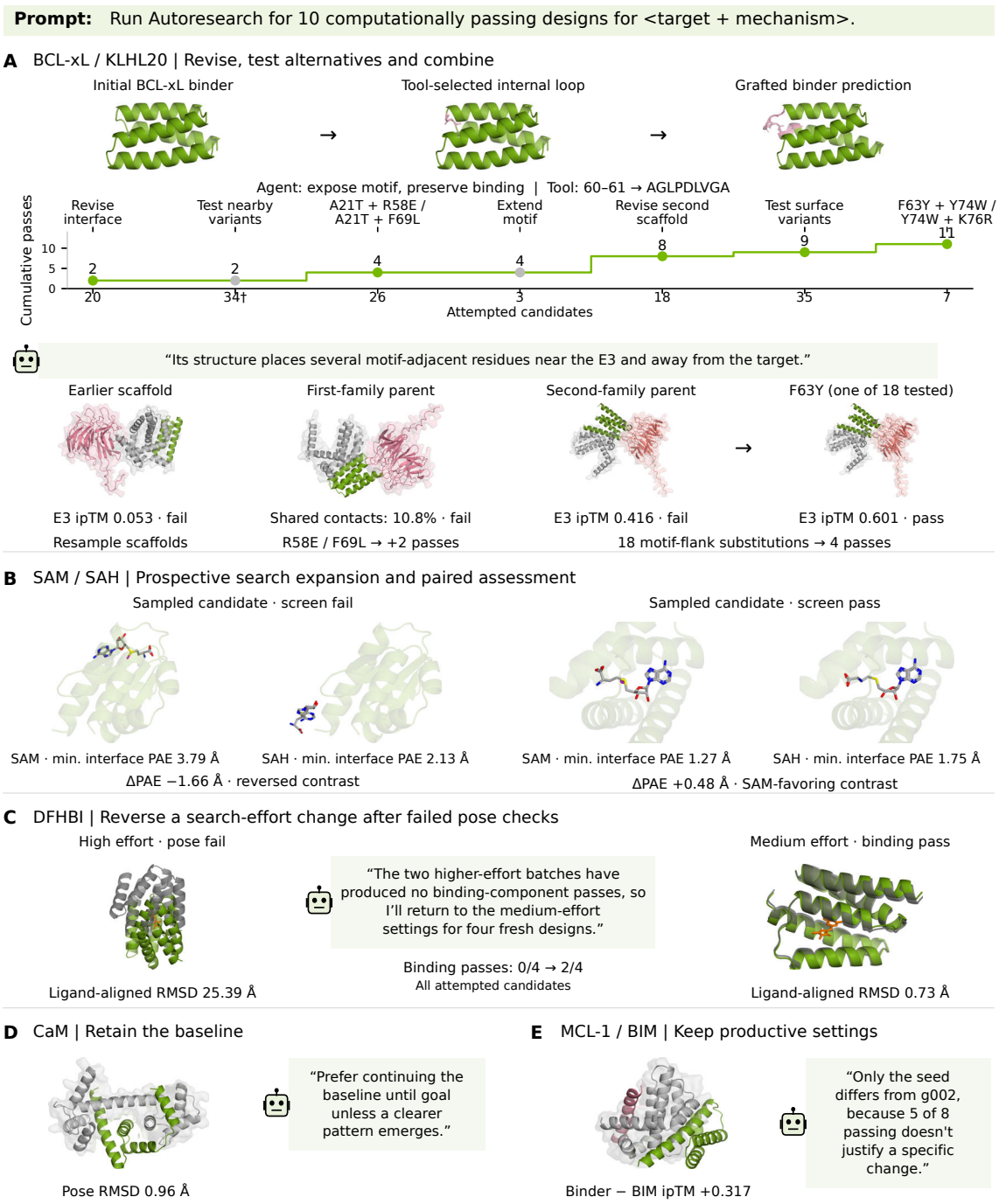


Figure 5 | From the initial goal to successive research decisions. (A) Internal-loop grafting of a KLHL20-recruiting motif (Mylemans et al., 2026; Chen et al., 2019), followed by tested revisions. F63Y was one of 18 tested substitutions. Counts below the curve include one unresolved submission (†), not a scientific failure. Pink marks the replaced loop or inserted motif. (B) Prospective search expansion and paired SAM/SAH assessment of different sampled designs. (C) DFHBI binding passes increased from 0/4 to 2/4 attempted candidates. The high-effort denominator includes one invalid prediction. Representative structures show the generation reference (gray) and independent prediction (green). (D,E) Recorded decisions to retain CaM and MCL-1 settings. Targets are silver, designs green and partners pink. SAM's extra methyl carbon is magenta.

2.5. Designed binders preferentially recognize calcium-loaded calmodulin

In a human-guided BindClaw campaign, we used paired-state computational comparisons to select designs, then shipped designs for experimental testing of whether they preferentially bound calcium-loaded (holo) CaM over the calcium-depleted (apo) condition. The design campaign combined evidence-guided binding sites, a holo template, RFD4-Proteina generation with AF2-guided selection, and paired-state computational screening mimicking wet-lab assays (Fig. 6). Of 1,553 evaluated designs, 351 passed the initial in silico screen. Final filtering and selection yielded 18 unique sequences for experimental testing (Fig. 6A). Exact filters and the reconstructed selection procedure are described in App. C.

Binding was measured by biolayer interferometry (BLI) through Adaptyv Bio, comparing calcium-loaded CaM (2 mM $CaCl_2$) with calcium-depleted CaM (3 mM EDTA). Holo BLI measurements included a long dissociation phase of approximately 1,200 s. Fifteen of 18 submitted designs showed holo binding, while two had unassessable loading. Primary affinities are BLI replicate medians, with replicate variability reported in App. C. Among the holo binders, 13 showed no detected apo binding. The two with quantified apo binding, #13 and #09, had 72.8- and 42.5-fold lower median apparent K_D for holo CaM, respectively. SPR measurements under calcium-loaded conditions provide additional binding evidence and are reported separately, without averaging across platforms (Table 5).

Design #10 illustrates holo binding and apo nondetection (Fig. 6D,E). Its BLI apparent K_D estimate is 151 pM, with precision limited by the fit reaching the imposed lower bound for k_{off}. Design #15 had a median BLI apparent K_D of 254 pM (replicate range 252–257 pM). For design #10, interface confidence values computed from Protenix-predicted structures of the holo complex were higher than those of the apo complex, with ipTM of 0.909 and ipSAE of 0.697 for holo, compared to 0.340 and 0.000 for apo, respectively (Zhang et al., 2026; Dunbrack, 2025) (Fig. 6C). Metrics derived from predictions through Boltz-2 gave the same direction of preference, with holo/apo ipTM values of 0.913 vs. 0.644 and ipSAE values of 0.790 vs. 0.176 (Passaro et al., 2025). AlloGen's mean predicted-pose Q score across three seeds was 0.897 for holo, and 0.004 for apo, giving a holo-minus-apo contrast of $+0.894$ (Cao et al., 2026). These paired scores consistently favor the holo-conditioned complex, in agreement with the observed holo binding and apo nondetection. ESM3ΔG also predicted the designed binder to be stable (6.41 kcal/mol) (Cho et al., 2026).

3. Conclusion

Augmenting in silico generation with mechanistic insights may require considering the native interaction for an inhibitor, the recruited partner for an assembly, or the alternative state for a state-selective binder. BindClaw connects these biological requirements to protein generation backends, and adopts assessment informed by biological function. Furthermore, agents within the assay builder can construct novel computational assays from supplied evidence, while agents within the autoresearch loop use prior evidence to guide subsequent design campaigns.

Our studies examine BindClaw from complementary perspectives. A nine-target benchmark showed that the best-performing binding site can differ between protein design backends, and BindClaw can select suitable sites for each backend. Autoresearch agents prompted helical conditioning, a narrower length range and more sampling after observing weak IL-2/IL-15 receptor predictions (Fig. 4F). For BCL-xL/KLHL20, weak E3 engagement prompted agents to introduce motif-flanking mutations. After F63Y passed, the agent tested combinations of individually passing mutations (Fig. 5A). These cases illustrate how traces from previous campaigns help agents take decisions toward achieving the design objective. In a human-assisted CaM design campaign, 15 of 18 BindClaw

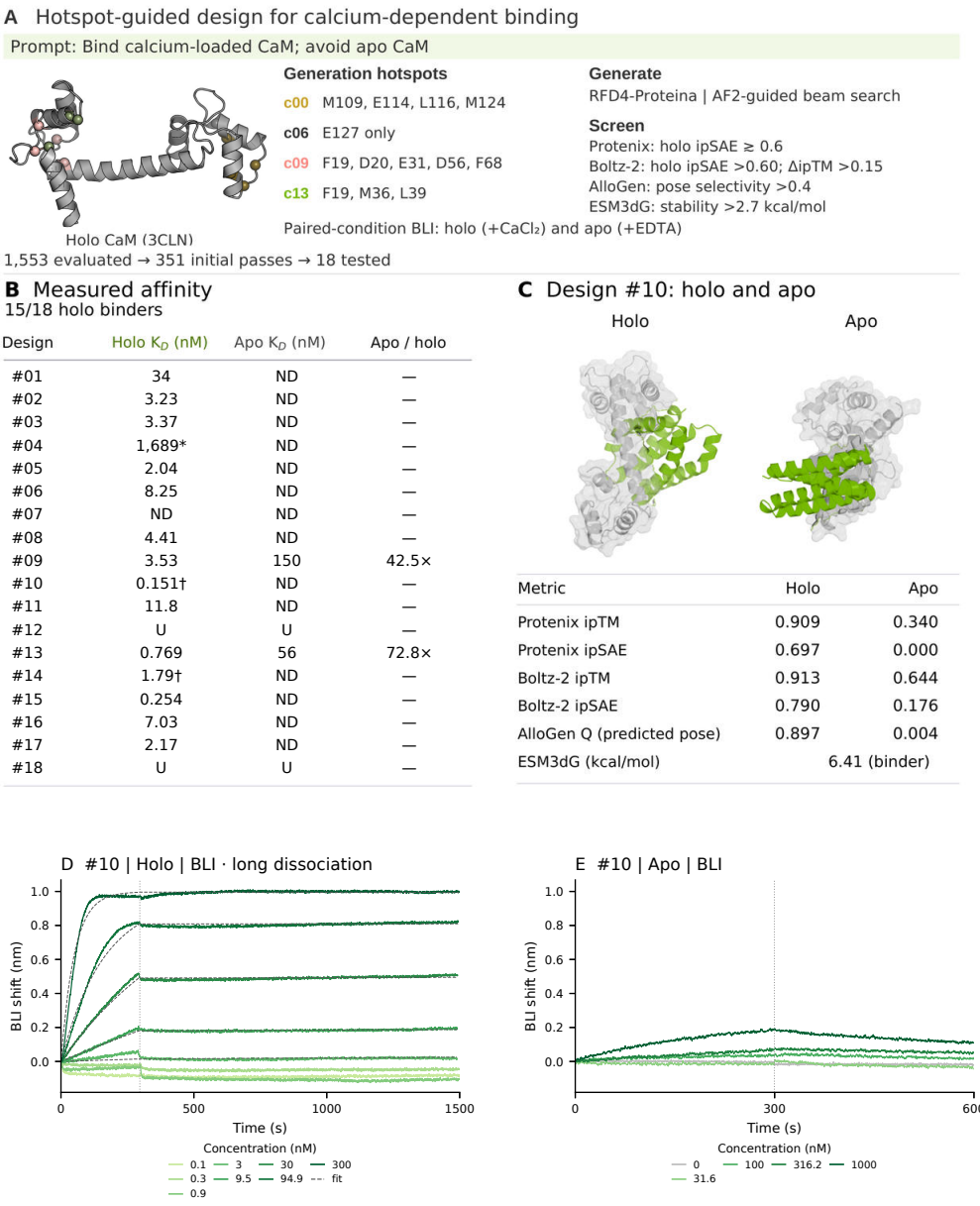


Figure 6 | Calcium-condition-dependent design and measured binding. (A) Hotspot inputs and selection of 18 tested sequences. (B) BLI binding calls and replicate-median apparent K_D values in nM. ND denotes a nonbinder classification and U unassessable loading. $*$ marks an estimate beyond the tested concentration range. † marks a contributing rate-bound fit. Replicate ranges and separate SPR affinities appear in App. C. (C) Design #10 predictions, with CaM in silver and binder in green. (D,E) Design #10 BLI measurements under holo and apo conditions, on matched response scales. The holo BLI apparent K_D estimate is 151 pM. The fit reaches the imposed lower bound for k_{off}. Dashed curves show the original selected holo 1:1 fit without refitting. Dotted lines mark dissociation onset.

designs bound calcium-loaded CaM (83%). BLI estimates included three subnanomolar and nine single-digit nanomolar apparent K_D values, with the estimate for #10 limited by the imposed off-rate bound. SPR measurements provided additional binding evidence, with affinities reported separately. Together, these results show a high binding yield and a consistent preference for calcium-loaded CaM among the tested binders.

However, BindClaw relies on computational evaluators whose accuracy varies across targets and molecular contexts. If an oracle cannot reliably predict a relevant interaction, conformational change or molecular assembly, its scores cannot reliably guide design toward that objective. Computational comparisons of partners, states and controls mimic selected aspects of the intended biological experiment, but do not establish biological activity. In particular, they do not fully address cellular processes such as endocytosis, internalization, intracellular delivery and access to targets in different cellular compartments. Extending BindClaw to these objectives will require additional computational tools and assays that measure the relevant cellular steps.

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A. Methods

Define the biological question.

- **Specify the objective and inputs.** The user provides a desired function and available sequences, structures, papers or target identifiers.
- **Define the comparison.** BindClaw identifies the experimental construct, relevant partners or alternative states, and controls.
- **Propose binding sites.** Literature and structural annotations nominate sites that are mapped onto the construct. Multiple hypotheses can be tested because agreement among sources does not establish design success.
- **Build the assay.** The assay builder packages supplied documents using Paper2Agent conversion scripts (Miao et al., 2025). Agents can retrieve additional literature through PaperClip (GXL, 2025). BindClaw constructs the assay and tracks provenance.

The assay preserves the biological objective and required comparisons throughout the campaign.

Represent the molecules and generate candidates. Natural proteins are evaluated with verified homolog MSAs or valid, construct-matched structural templates. State-dependent comparisons retain the required state templates, and native-partner competition retains its prescribed sequence conditioning. Designed proteins are evaluated as single sequences without MSAs or templates. BindClaw connects these inputs to supported generators, including RFD4-Proteina (Corley et al., 2026), Proteina-Complexa (Didi et al., 2026) and Protein Hunter (Cho et al., 2025b). Each comparison records the generator, molecular inputs and allowed changes. The biological target and assessment are held fixed when comparing design choices.

Assess binding and the requested mechanism. Predicted structures first establish whether the candidate meets the recorded structural criteria. A separate, mechanism-specific comparison then examines the relevant site, partner, state or assembly (Fig. 2). For example, a candidate proposed to block a native interaction is assessed with the native partner present. A state-selective candidate is compared across the specified states. Controls provide reference measurements for interpreting those comparisons. Incomplete controls and unqualified comparisons are reported separately from candidate outcomes.

Revise designs and deliver the evidence. The coding agent can propose new sites, generator settings or sequences within the permitted budget and available tool interfaces. It retains the evaluator while comparing these changes, records unsuccessful attempts, and preserves the relationship between parent and revised designs. A change to the assay creates a new assessment version. Optional repeat predictions assess computational repeatability. The final report links candidate sequences and structures to controls, comparison results, recorded revisions and unresolved requirements, so that experimental follow-up can target the stated biological question.

A.1. Generation settings, models and evaluation criteria

Input records. Generation and evaluation recipes are separate, content-hashed artifacts. They record modality, design operation, mechanism, exact constructs, chain origins, partners, states, source mappings, controls and unsupported endpoints. Evaluation inputs are prepared separately from generation inputs, preserving ligand chemistry, modifications and chain identity.

Current generation and assessment.

- **RFD4-Proteina search.** Native medium search uses 60 steps, four beams and four branches. AF2 guides de novo protein search at step 60. Native OpenFold3 guides other supported modalities at steps 40 and 60. AF2 initial guesses are disabled, and only natural chains receive templates.
- **Rollout retention.** For protein-target search, saved rollouts meet native AF2 $\mathrm{iptm_coi} \geq 80/100$. For other supported target modalities, OpenFold3-guided rollouts meet minimum interface PAE≤ 1.5 Å. These criteria retain candidates for assessment rather than establish final qualification.
- **Other generators.** Proteina-Complexa uses its unguided structured route. Protein Hunter alternates Boltz-2 prediction with sequence redesign (Cho et al., 2025b; Passaro et al., 2025).
- **Supported inputs.** RFD4-Proteina and Protein Hunter adapters cover protein, ligand, DNA and RNA. Proteina-Complexa covers protein and ligand.
- **Assessment models.** The current map uses Protenix-v2 for de novo protein targets and OpenFold3 for supported other modalities. Frozen cohorts retain their recorded map.
- **Model independence.** Reusing the search model family is a separate execution, not confirmation by an independent model family.
- **Model terms.** Source-code licenses and checkpoint-use terms are separate.

Models, tools and recorded versions. Table 1 maps models and tools to their roles, recorded versions, references and source repositories.

Table 1 | Model and tool usage, recorded versions and source attribution. GitHub links identify upstream sources. Local modifications and study-specific settings are described separately.

Model / tool	Role and study scope	Recorded version / checkpoint	Source repository
Paper2Agent (Miao et al., 2025)	Document conversion and evidence packaging for the assay builder.	Vendored converter source c5ce59cc726e. Upstream scripts are unchanged. BindClaw supplies assay construction.	jmiao24/ Paper2Agent
Paperclip (GXL, 2025)	Literature search and full-text extraction interface.	Exact study package version not recorded in the inventory.	GXL-ai/ paperclip
Proteina-Complexa (Didi et al., 2026)	Generation with standard and beam configurations in the nine-target site benchmark.	Inventory: registered source 5ae24b055d82, separate checkout abf53e953640, with local modifications. Exact cohort weight receipt unresolved.	NVIDIA-BioNeMo/ Proteina-Complexa
RFD4-Proteina (Corley et al., 2026)	Generation and guided search for the site benchmark, CaM and mechanism-directed campaigns.	Inventory source de864178d28e, flow epoch 169 / step 340000 EMA, autoencoder 4 September 2026, z16 / 400k EMA. Recorded runs retain their own records.	Rosetta Commons/ RFD4-Proteina
Protein Hunter (Cho et al., 2025b)	Iterative structure prediction and sequence redesign for the site benchmark.	Local modified implementation. Boltz package 2.2.1 in the inventory. Exact cohort source/weight receipt unresolved.	yehlincho/ Protein-Hunter

Model / tool	Role and study scope	Recorded version / checkpoint	Source repository
PXDesign (Team et al., 2025)	Fourth generator family in the nine-target site benchmark.	Exact cohort source revision and checkpoint not established by the software inventory.	bytedance/PXDesign
AlphaFold2 / Multimer (Jumper et al., 2021; Evans et al., 2021)	RFD4-Proteina search guidance, including CaM.	Recorded assets: model_1_ptm and model_1_multimer_v3. SHA-256 prefixes 5e564f79af5b, 611da8fc7478. The exact choice is run-specific.	google-deepmind/alphafold
Protenix-v2 (Zhang et al., 2026)	Common structural evaluator for the site benchmark, protein-complex assessment and mechanism assessment.	Inventory package 2.0.0, protenix-v2.pt, SHA-256 prefix 8f931f9774a39.	bytedance/Protenix
Boltz-2 (Passaro et al., 2025)	Protein Hunter prediction/redesign and paired holo/apo CaM screening.	Inventory package 2.2.1 in the Protein Hunter runtime. The exact CaM weight hash not established by that inventory.	jwohlwend/boltz
OpenFold3 (The OpenFold3 Team, 2026)	Structural prediction in the recorded autoresearch workflow and supported non-protein routes in the current evaluator map.	Inventory package 0.4.1, native of3-p2-155k.pt (preview-2). Converted guidance weights are a separate artifact.	aqlaboratory/openfold-3
RoseTTAFold3 (Corley et al., 2025)	Available structure-prediction interface in the architecture. Installation alone does not establish cohort use.	Foundry source 4010e3e2e735, rf3_foundry_01_24_latest_remapped.ckpt.	Rosetta Commons/foundry
AlloGen (Cao et al., 2026)	CaM state-selectivity scoring. Predicted-pose and redocked scores are kept distinct.	Inventory source 2327665e9c46, locally modified, checkpoint SHA-256 prefix 716e4716c001.	Local source. Public repository not verified
ESM3ΔG (Cho et al., 2026; Cho, 2025)	Binder stability annotation, including the CaM score table.	Inventory esm package 3.2.1.post1. This is not the stability-head checkpoint version. Exact ensemble weight receipt unresolved.	yehlincho/absolute-stability-predictor

Structural profiles and exact measurements.

Profile	Required measurements
Protein Gate 1	Binder-chain pTM ≥ 0.80, minimum interface PAE ≤ 1.5 Å and target-aligned binder-backbone RMSD ≤ 2.5 Å.
Ligand	ipTM ≥ 0.80, minimum interface PAE ≤ 1.5 Å, binder-backbone RMSD ≤ 1.5 Å and ligand RMSD ≤ 5 Å under the same binder alignment.
DNA	Minimum interface PAE ≤ 1.5 Å and target-aligned binder-backbone RMSD ≤ 8 Å.
RNA	Extends the DNA profile. It is not a separate source-protocol benchmark.

The cutoffs follow the supplied RFD4-Proteina protocol (Fig. 2, pp. 7–8). These adopted cutoffs are fixed structural heuristics, not calibrated operating points for these predictors. For binder tokens B, target-partner tokens T_k and directed PAE P, the implemented statistic is

$$\mathrm{minIPAE} = \min_k \tfrac{1}{2}\Big(\min_{i\in B, j\in T_k} P_{ij} + \min_{i\in T_k, j\in B} P_{ij}\Big).$$

It is neither mean interface PAE nor ipSAE. pTM is binder-chain-local. Confidence scales are explicitly normalized. RMSD uses exact identities and the hash-pinned generated reference: N/Cα/C/O (not OXT) for protein backbones, mapped heavy atoms for nucleic-acid targets and ligands. Missing metrics cannot be replaced by confidence aliases.

A.2. Assay builder for paper-derived objectives

IL-2 and IL-15 share two signaling receptor chains, IL-2Rβ and the common γ chain. Following Silva et al. (2019), the assay builder campaign aims to design a protein that engages this shared receptor pair (Fig. 4A,F). The example below follows the first two completed design batches shown in panel F. The first box quotes recorded campaign instructions, rather than a verbatim original user message. The remaining boxes summarize saved assay, generation and evaluation artifacts. They are not reconstructed dialogue.

Recorded campaign instructions | verbatim excerpt

"Design non-antibody IL-2/IL-15 mimics preserving the paper's receptor engagement objective. Recover exact Neo-2/15 constructs and IL-2Rβ/common-γ partners, compare productive dual engagement against missing/reoriented partner controls and relevant receptor-selectivity alternatives."

Constructed computational assay | saved specification

Use the mouse IL-2Rβ/common-γ assembly in 6DG5 as the structural reference. For each predicted ternary complex, measure candidate-interface ipTM and contacting residues separately for the two receptors. Align IL-2Rβ to the reference and measure the common-γ Cα RMSD to assess receptor arrangement. The recorded structural criteria require ipTM ≥ 0.50 and at least ten contacting residues at each interface, with receptor-arrangement RMSD ≤ 4.0 Å. These are computational screening thresholds, not experimental activity thresholds from the source paper.
Independent OpenFold3 predictions use structural templates for the natural receptors and single-sequence inputs for the designed protein.

Initial design output | eight generated and evaluated candidates

The initial generation length range was 83–124 residues. Among these eight candidates, the design with the largest minimum receptor-interface ipTM had IL-2Rβ/common-γ ipTM values of 0.074/0.192 and a receptor-arrangement RMSD of 75.49 Å. This is the 113-residue candidate shown on the left of Fig. 4F. Its helical fraction was 0.811.

Agent decision | revised generation settings

After the weak dual-receptor predictions and the occurrence of non-helical candidates, the agent added helical conditioning, narrowed the generation length range to 90–110 residues and increased the next batch from eight to 24 candidates. The prediction model and structural comparison were retained.

Next design output | 24 generated and evaluated candidates

The 96-residue candidate shown on the right of Fig. 4F had IL-2Rβ/common-γ ipTM values of 0.859/0.904, 14/21 contacting residues and a receptor-arrangement RMSD of 3.97 Å. Its helical fraction was 0.936. The displayed candidate in each batch was selected by the largest minimum of the two receptor-interface ipTM values.
The workflow retained candidate sequences, predicted complexes and per-candidate measurements for subsequent analysis.

Scope and provenance. This example shows how the agent used predicted receptor engagement and geometry to revise generation settings. The two-batch comparison is a recorded search trajectory. The original pipeline left full binding eligibility unresolved, and the reference prediction did not recover the required receptor arrangement. The scores therefore describe structural candidates for

further testing, without establishing cytokine signaling or potency. Missing-partner and accessory-receptor diagnostics were run subsequently and are separate from the two batches shown here.

B. Hotspot evidence, ablation and model sensitivity

Reading the hotspot comparison. Figure 3 covers nine targets, six hotspot sets and six configurations from four generator families (6,480 valid evaluations). The main figure reports percentages out of 20 valid evaluations per cell. The complete supplementary heatmap displays passing counts on a common 0–20 scale. The frozen input table retains descriptive Wilson intervals. These do not model shared-search dependence. Validity-only replacements and unequal compute preclude efficiency comparisons.

Source provenance for the common-evaluator reevaluation. Table 2 gives all 54 recorded shared patches, including PDB author residue identifiers and source channels. All 640 patch-residue memberships match identities in the nine hash-verified saved target structures. Source labels and literature identifiers remain recorded annotations, not independently verified biological evidence.

Table 2 | **Exact shared hotspot inputs.** PDB author numbering. Chain prefixes are retained. Set numbers are target-specific. Sources: D, deterministic. A, structural Analyst. L, LLM search. P, Paperclip. I, interface. Saved coordinate hashes and residue identities were verified. Backend-specific transformations and full-sequence mappings remain unaudited.

Target / PDB	Set	Source	Residues (saved order)
CDK4 / 2W96	1	D	B140, B132, B135, B57, B134, B133, B59, B55, B58
	2	A	B9, B10, B12, B30, B13, B97, B96, B98
	3	L	B12, B13, B20, B14, B19, B33, B97, B15, B98, B95, B96, B35, B18, B99, B17, B147, B16, B93, B94, B100, B73, B140
	4	P	B20, B35, B158, B24, B147, B61
	5	I	B41, B42, B48, B49, B50, B43, B82, B44, B81, B51, B53, B54, B55
	6	D	B24, B76, B35, B77, B73, B78, B79, B61
EGFR / 3IKA	1	D	A719, A724, A745, A746, A747, A855, A748, A787, A858
	2	A	A718, A719, A720, A794, A795, A855, A838
	3	L	A718, A719, A745, A793, A724, A723, A855, A856, A837
	4	P	A718, A726, A743, A796, A745, A792, A797, A793, A844, A790, A791, A723, A747, A788, A854, A855, A775, A777
	5	I	A704, A705, A706, A780, A763, A707, A781, A760, A782, A761, A787, A708, A786, A783, A756, A784
	6	D	A837, A838, A835, A834, A833, A869, A861, A873
KRAS / 4DSN	1	D	A13, A14, A15, A60, A16, A34, A10, A17, A59, A35, A18, A61, A116, A117, A58, A19, A29, A57
	2	A	A12, A32, A33, A36, A37, A31, A38, A30, A40
	3	L	A5, A7, A56, A71, A40, A39, A72, A57, A58, A38, A20, A37, A16, A19, A17, A10, A36, A35, A15
	4	P	A5, A54, A55, A7, A74, A56, A41, A71, A40, A39, A70, A58, A38, A37, A16, A10, A36, A35
	5	D	A146, A147, A145, A28, A22, A152, A119, A30
	6	A	A64, A68, A69, A99, A96, A95
MCL1 / 3PK1	1	D	A231, A234, A227, A249, A267, A250, A266, A248, A253, A252, A251, A263, A254
	2	A	A224, A228, A270, A266, A231, A267, A263, A235, A253, A250, A249, A254
	3	L	A174, A175, A311, A179, A307, A312, A306, A310, A181, A304, A313, A182, A305, A309, A308, A314, A209, A315, A316, A264, A261, A268, A265, A317, A259, A260, A267, A318, A263, A262, A266
	4	P	A216, A220, A265, A266, A267, A263, A270, A224, A271, A228, A274, A227, A294, A250, A253, A231
	5	I	A215, A216, A220, A319, A320, A265, A266, A318, A262, A317, A261, A264, A267, A263, A260
	6	D	A260, A261, A262, A264, A318, A319

Target / PDB	Set	Source	Residues (saved order)
P53 / 2VUK	1	D	A168, A169, A167, A166, A165, A171, A170, A163, A164, A249, A162, A172, A173, A251, A250, A97, A246, A96, A252, A245, A161, A98, A174, A213, A253, A271, A160, A175, A214, A254, A270
	2	A	A174, A175, A214, A213, A179, A178, A181
	3	L	A120, A121, A280, A277, A281, A276, A283, A282, A275, A274, A273, A132, A241
	4	P	A145, A230, A147, A220, A157, A223, A222, A228, A150, A151, A152, A153
	5	I	A96, A97, A170, A169, A99, A168, A166, A167, A101
	6	D	A179, A178, A180, A182, A181, A176, A177, A238, A237, A191, A192, A183, A184, A242, A194, A193, A244, A243, A137, A138, A236, A190, A240, A195, A235, A274
PCSK9 / 2P4E	1	D	A153, A239, A157, A237, A391, A443
	2	A	A153, A154, A155, A238, A379, A237, A378, A377, A194
	3	L	A153, A154, A155, A238, A237, A243, A378, A195, A377, A372, A194
	4	P	A153, A238, A369, A379, A381, A237, A380, A378, A377, A194
	5	D	A226, A186, A386, A253, A374
	6	A	A192, A376, A374
PDL1 / 4ZQK	1	D	A20, A122, A121, A123, A120, A124, A117, A125, A54, A56, A66
	2	A	A54, A56, A115, A121, A122, A123, A58, A113, A124, A125
	3	L	A54, A117, A116, A118, A56, A115, A121, A122, A123, A58, A113, A124, A125
	4	P	A18, A19, A20, A121, A122, A123, A116, A115, A124, A54, A125, A56, A113
	5	I	A18, A19, A20, A120, A119, A121, A122, A21, A117, A123, A115, A124, A23, A54, A114, A55, A56, A113
	6	I	A26, A126, A125, A58, A63
TREM2 / 5ELI	1	D	A68, A67, A66, A47, A48, A77, A76, A88, A62, A87, A38, A36, A96
	2	A	A38, A43, A87, A44, A86, A48, A46, A47, A50, A67, A76, A77
	3	L	A38, A43, A42, A88, A87, A44, A48, A46, A47, A50, A66, A74, A70, A71, A67, A72, A68, A69
	4	P	A41, A44, A89, A87, A47, A70, A86, A71, A74, A69, A67, A68, A75
	5	I	A67, A68, A69, A72, A75, A77, A76, A78, A79
	6	D	A27, A28, A130, A31
WEE1 / 5V5Y	1	D	A328, A312, A307, A463, A465, A342, A431, A426
	2	A	A313, A314, A328, A374, A463, A379, A380, A433, A363, A435
	3	L	A306, A307, A313, A314, A308, A328, A326, A327, A310, A378, A431, A463, A433, A374, A373, A465
	4	P	A309, A310, A313, A328, A465, A463, A464, A426, A326, A376
	5	L	A342, A426, A425, A423, A422, A359, A420
	6	P	A377, A378, A379, A433

B.1. Evidence and construct mapping

Hotspot proposals combine abstract-level PubMed and Semantic Scholar retrieval, UniProt residue features, STRING partner information, PDB or AlphaFold structures, and Paperclip full-text extraction (Sayers et al., 2025; Kinney et al., 2023; Consortium, 2015; Szklarczyk et al., 2023; Burley et al., 2019; Varadi et al., 2024; GXL, 2025). These channels may share underlying experiments. Multiple votes do not establish independent evidence. Candidate positions must map to the chosen construct.

The structural tools estimate exposure from C_α neighbors, characterize hydrophobic/polar/charged patches and examine crystallographic B factors or AlphaFold confidence. The Analyst guidance requested 4–12 accessible, spatially clustered residues with biological relevance, avoiding highly flexible positions. Guides used both decile-based and top-five flexibility rules. The executed configuration and saved residue set determine the actual conditioning. The CaM campaign's distinct clustering and selected inputs are specified in Appendix C.

B.2. LLM comparison

Ten profiles each ran three repeats on six targets: MCL-1, CDK4, KRAS, MDM2, HER2 and PD-L1. Each trial used the native Analyst, HotspotSearcher, PaperclipSearcher and deterministic consensus stages. Profiles interpreted the same frozen corpus of 47 papers, prepared with Claude Opus 4.7. Other evidence tools ran normally. No binder generation or folding was performed.

Analyst/HotspotSearcher/Paperclip round and output-token budgets were 6/16,000, 6/12,000 and 10/32,000. Request/trial limits were 600/3,600 s, with six concurrent trials. Analyst's fast fallback was disabled, although DeepSeek used a later native fallback twice. Consensus required matching chain, residue position and amino-acid identity.

Figure 7 reports exact-residue Jaccard overlap, $J(A, B) = |A \cap B|/|A \cup B|$. Available repeat pairs are averaged within each target, then equally across targets. Diagonals compare distinct repeats. Of 180 executed trials, 145 had complete artifacts and 139 had nonempty outputs in all four channels. Nemotron/Qwen agreement with Claude Opus 4.7 was 0.51/0.57. Shared deterministic evidence and serving differences prevent attributing these comparisons solely to model reasoning.

Offline replay reproduced all 180 consensuses. A parser correction recovered two Qwen outputs but was not substituted into the primary results.

Site location, size and geometry vary with the source labels, and collection rules were amended after inspection of earlier results. These comparisons do not isolate the causal contribution of a particular evidence source.

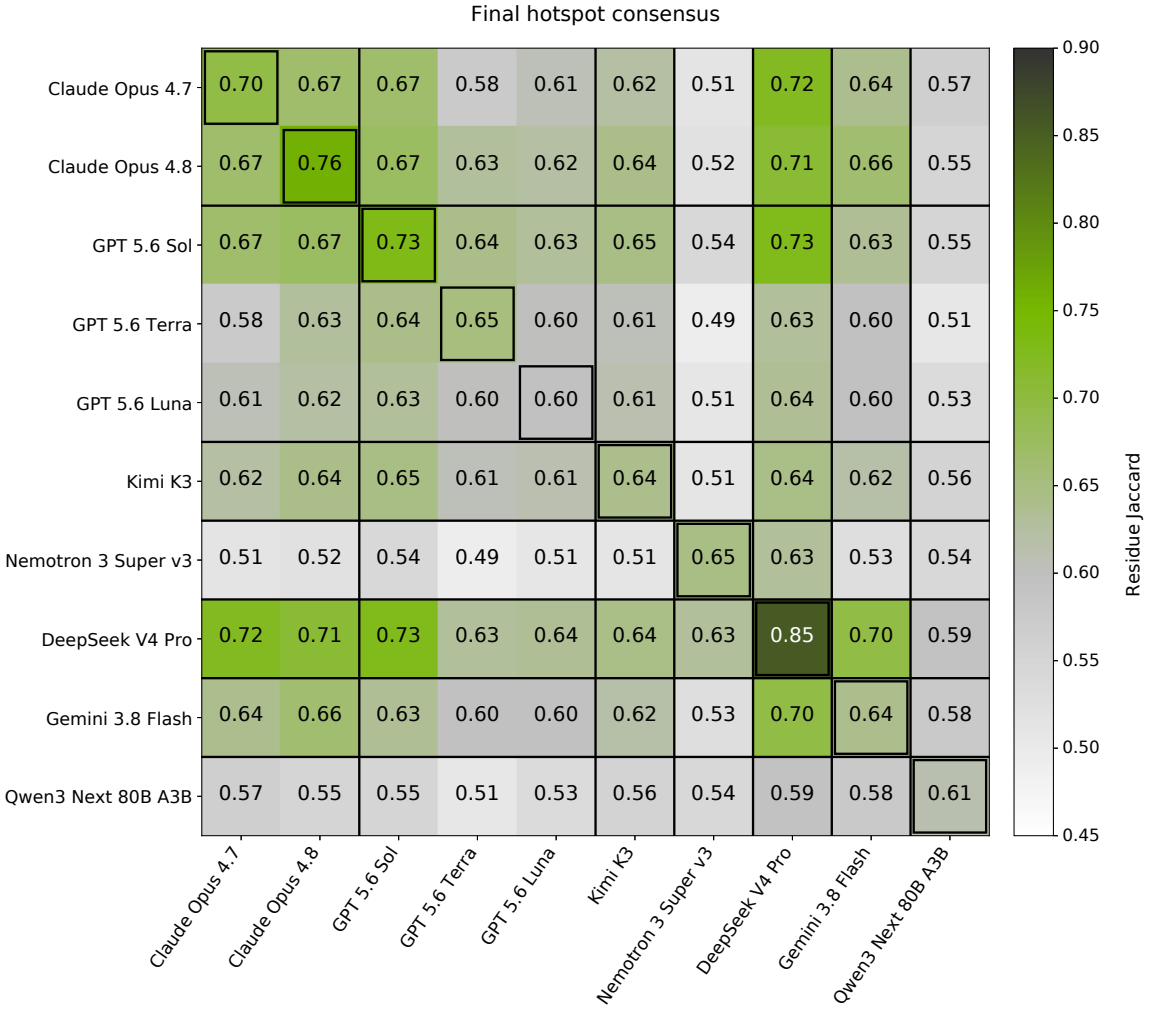


Figure 7 | Hotspot-consensus agreement across ten LLM profiles. Cells show exact-residue Jaccard overlap, averaged equally over available targets. Outlined diagonal cells compare distinct repeats of the same profile. Black lines separate model families. The color scale spans 0.45–0.90. Agreement does not measure experimental hotspot accuracy.

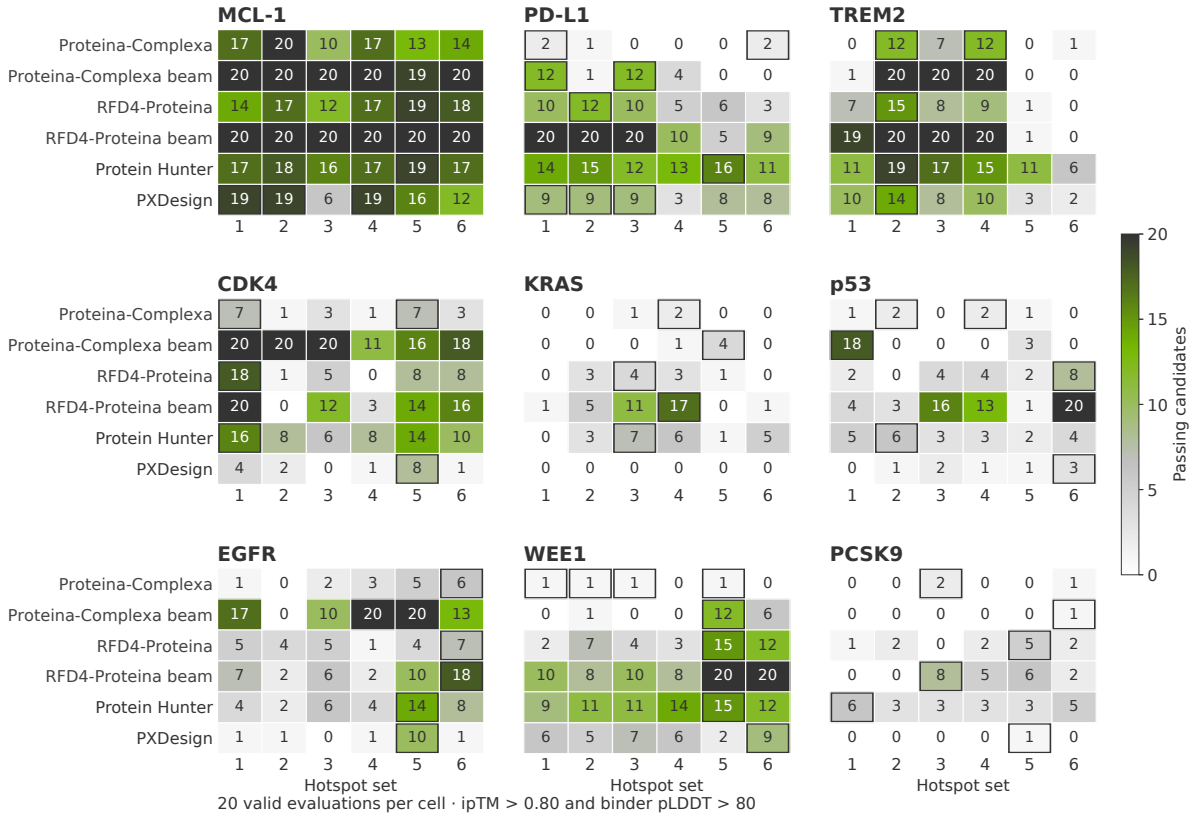


Figure 8 | **Complete target, site and generator comparison.** Nine targets, six hotspot sets and six generator configurations give 324 cells, each containing 20 valid evaluations (6,480 total). Passing requires ipTM > 0.80 and binder pLDDT > 80. Counts share a 0–20 scale. Positive maxima and ties are outlined within each target–model row.

C. CaM design and binding experiments

BLI measurements. Adaptyv Bio performed multi-cycle BLI using human CALM1 as soluble analyte. Designed binders carried GFP11/Twin-Strep tags, were expressed cell-free at 37 °C for 8 h and quantified by split-GFP complementation. Measurements used a Gator Bio instrument, Strep-Tactin XT probes, 25 °C and 5 Hz sampling. Apo buffer contained 10 mM HEPES, 150 mM NaCl, 0.2% Tween-20 and 3 mM EDTA, pH 7.4. Holo buffer replaced EDTA with 2 mM $CaCl_2$. The protocol specifies EDTA despite the source package's "+EGTA" name.

Association lasted approximately 300 s. Initial dissociation lasted 300 s, and follow-up holo measurements extended it to approximately 1,200 s to better resolve slow off-rates. The displayed holo measurement for #10 uses this longer window, while apo measurements retain 300 s dissociation. Concentration panels varied by measurement, with holo measurements extending down to 0.1 nM. Analysis used baseline/reference subtraction and global kinetic fitting, sharing rate constants across concentrations with per-curve response maxima. The default model was 1:1 Langmuir binding.

Design and selection. Under human guidance, BindClaw agents proposed binding sites from peptide contacts, calcium-associated residues and structural pockets. These proposals were clustered on 1CLL to define alternative sites for generation: c00/c06 covered C-lobe patches, c09 used PaperClip contact/calcium evidence, and c13 covered an N-lobe pocket. The four tested c06 designs were generated using E127 alone rather than the full six-residue parent patch. The campaign explored these site hypotheses with RFD4-Proteina, varying binder length (50–100 or 95–130 residues) and random seed. Generation used holo CaM coordinates from 3CLN (Babu et al., 1988), C_HOT selectors, C_SEC:HELIX conditioning and AF2 wide-beam search, retaining up to 20 candidates per task. Paired holo/apo screening then selected candidates for the final computational filters and experimental testing.

After removing repeated scoring rows, 1,553 design IDs remained. Initial requirements were monomer scRMSD < 3 Å, holo ipTM > 0.8, holo ipSAE > 0.45 and holo-minus-apo ipTM > 0.2. Of 351 initial passes, three lacked sequences, leaving 348 final metric records. The denominator counts scored IDs rather than unique sequences.

All 18 selected designs had Protenix holo ipSAE ≥ 0.6, Boltz-2 holo ipSAE > 0.60, Boltz-2 holo-minus-apo ipTM > 0.15, Protenix/Boltz-2 complex RMSD < 2.5 Å, predicted-pose AlloGen selectivity > 0.4, scRMSD < 1.1 Å, ESM3dG > 2.7 kcal/mol, glutamate fraction < 0.30, pI outside [6.5, 7.5] and radius of gyration < 17 Å. Final filtering and selection yielded 18 unique sequences for experimental testing (Fig. 6A). Predicted-pose and redocked AlloGen scores remain separate. Exact ranking weights were not reconstructed.

Binding outcomes. Table 4 retains binding calls and reports platform-specific BLI replicate medians without refitting. Replicate fits with binding reported and fit quality rated Good or Medium contribute to the arithmetic median and observed range. Numerical fit-quality cutoffs were not supplied. ND denotes a nonbinder classification, not a numerical affinity bound. Designs #12 and #18 had unassessable loading. The sole reported BLI fit for #10 gives an apparent K_D of 151 pM but reaches the imposed bound $k_{off} = 10^{-5}\,s^{-1}$, limiting its precision. Design #04 has a BLI median apparent K_D of 1,689 nM (range 617–3,529 nM), with estimates extending beyond the tested concentration range. One BLI replicate of #14 also reaches the imposed bound and is flagged.

Apo BLI measurements for #09 and #13 each had two quantified replicates, with median apparent K_D values of 150 nM (range 139–161 nM) and 56.0 nM (range 51.9–60.0 nM), respectively.

Table 3 | **RFD4-Proteina computational outcomes by actual hotspot input.** Strong holo: Protenix ipTM > 0.8. Paired: additionally holo-minus-apo ipTM > 0.2. All four: additionally holo ipSAE > 0.45 and scRMSD < 3 Å. Positions use 3CLN author numbering. Adaptive sampling differs between groups.

Group	Actual hotspot positions	Evaluated IDs	Strong holo	Paired	All four
c00	109/114/116/124	490	464	226	110
c06 (E127)	127	301	261	160	90
c06 (full)	127/129/131/140/141/144	93	32	18	11
c09	19/20/31/56/68	222	181	108	63
c10	51/67/71/72	199	187	99	40
c11	53/68/71/72	34	33	13	2
c12	129/133/140	96	8	6	3
c13	19/36/39	78	75	37	32
c14	102/132/134	40	0	0	0
Total		1,553	1,241	667	351

SPR measurements under calcium-loaded conditions are summarized independently in Table 5. BLI and SPR values are not pooled. The M13 positive control has a median SPR apparent K_D of 3.65 nM (range 3.40–3.90 nM), with non-standard kinetics that warrant caution in interpreting the exact value. Supplied conservative BLI-only interval summaries are shown separately in Table 6, rather than being relabeled as intervals for the arithmetic medians.

C.1. Response curves and displayed predictions

Figure 9 displays the first numbered BLI series with the longest available dissociation window for each design and condition. Series were not selected by affinity or fit quality. The main figure uses #10 BLI series 1 in both conditions, with the original selected 1:1 holo fit overlaid. Figure 10 shows the one holo and two apo BLI series in the current dataset. SPR curves and selected fits are displayed separately in Fig. 11. Plots use supplied processed responses without additional smoothing, refitting or baseline correction. BLI response is in nm and SPR response is in RU.

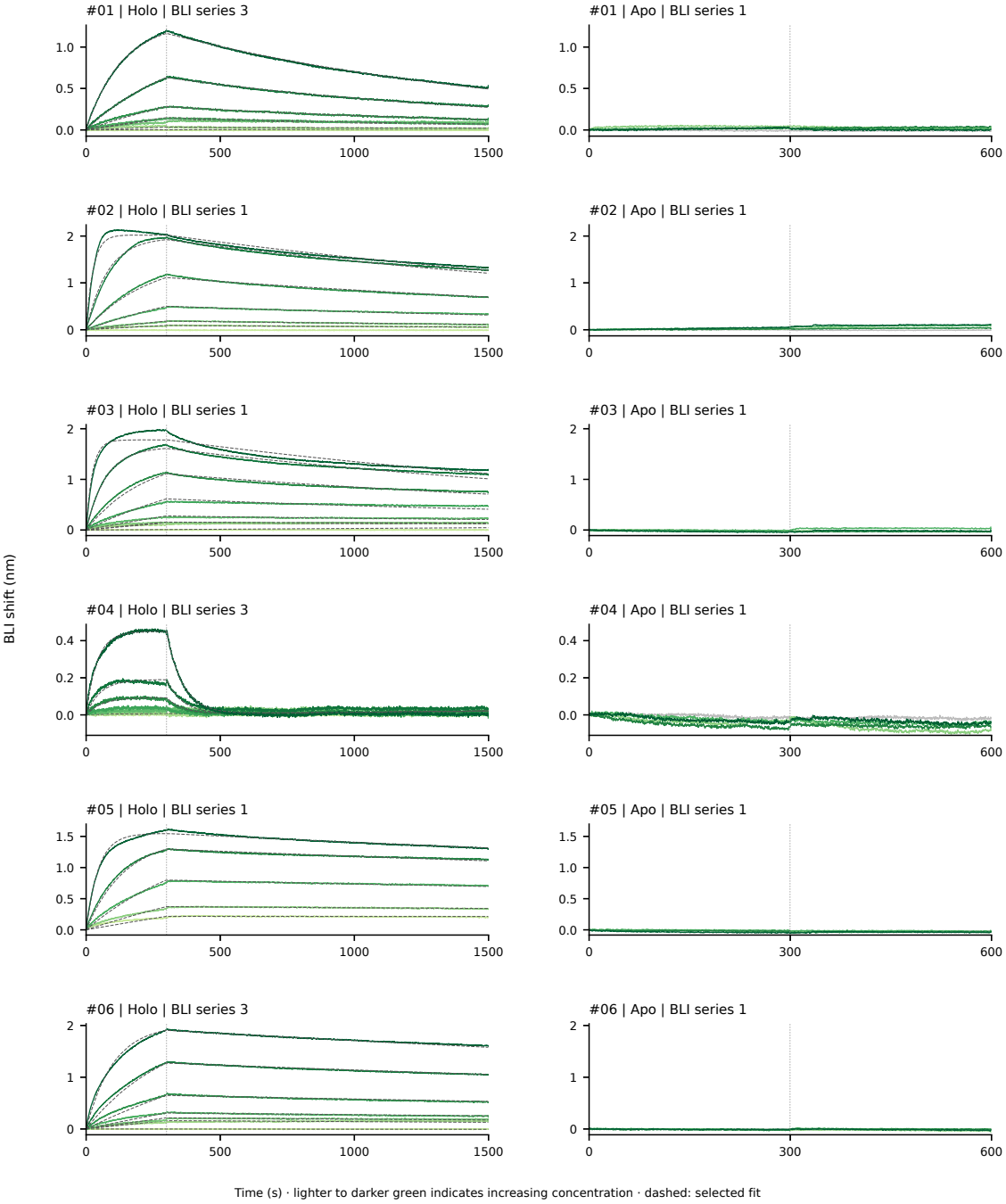


Figure 9 | **BLI curves for 16 assessable CaM designs.** The first numbered BLI series with the longest available dissociation window is shown for each design and condition. Lighter to darker green denotes increasing concentration, dashed lines show supplied selected fits, and dotted lines mark dissociation onset. Responses have not been refitted or smoothed. The holo window extends to approximately 1,200 s where available, while apo dissociation is 300 s. Designs #12 and #18 are omitted because loading was unassessable.

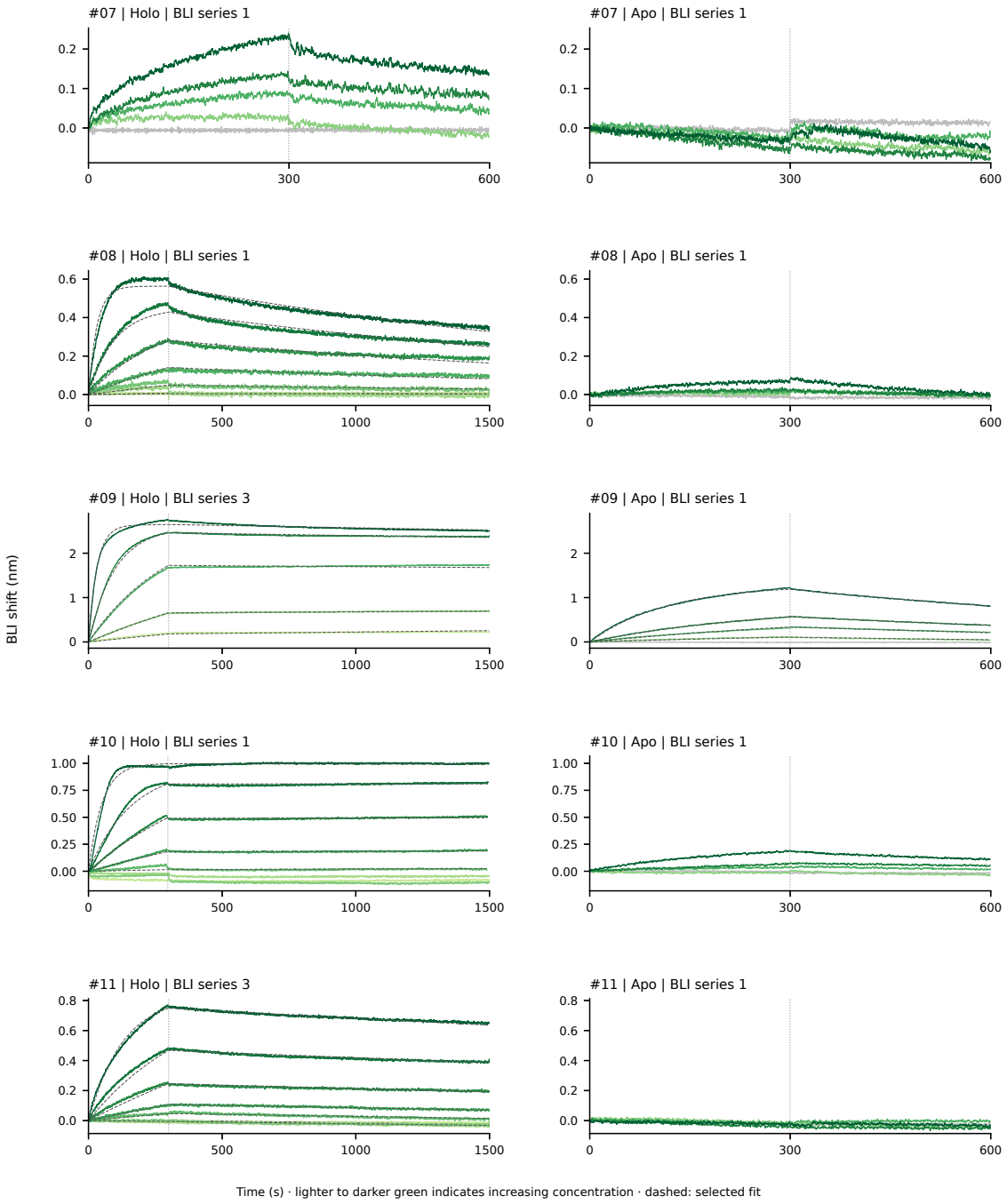


Figure 9 | **BLI curves for 16 assessable CaM designs (continued).** Series selection, colors and processing follow the preceding page.

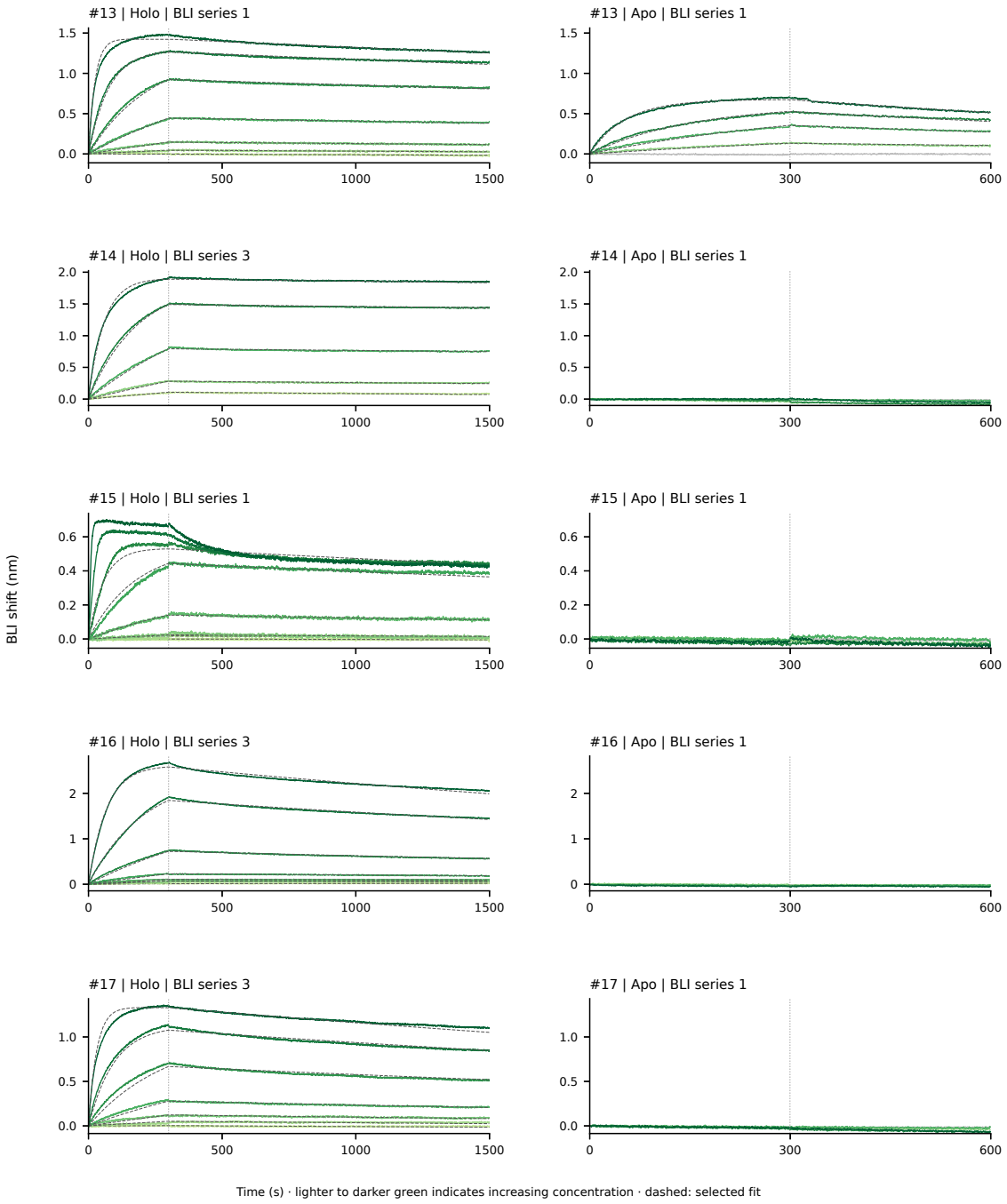


Figure 9 | **BLI curves for 16 assessable CaM designs (continued).** Series selection, colors and processing follow the preceding page.

Table 4 | **Computational scores and BLI outcomes for all 18 RFD4-Proteina designs.** Designs are numbered consecutively after excluding PH designs. Apparent K_D is the BLI replicate median in nM. ND: nonbinder classification. U: unassessable loading. Δ: holo-minus-apo ipTM. AlloGen columns distinguish predicted-pose and redocked scores. $*$: estimate extends beyond the tested concentration range (#04). †: one contributing replicate reaches the imposed off-rate bound.

Design	Group	scRMSD (Å)	PX Δ	B2 Δ	AlloGen pred.	AlloGen redock	Holo K_D	Apo K_D	Apo/holo
01	c09	0.395	0.676	0.789	0.906	0.879	34	ND	–
02	c00	0.602	0.557	0.755	0.517	0.514	3.23	ND	–
03	c06	0.406	0.593	0.676	0.944	0.943	3.37	ND	–
04	c06	0.304	0.593	0.346	0.987	0.998	1,689*	ND	–
05	c13	0.744	0.590	0.472	0.850	0.710	2.04	ND	–
06	c09	0.876	0.477	0.451	0.579	0.576	8.25	ND	–
07	c09	0.533	0.567	0.703	0.556	0.552	ND	ND	–
08	c09	0.363	0.578	0.522	0.992	0.057	4.41	ND	–
09	c09	0.316	0.657	0.291	0.944	0.945	3.53	150	42.5
10	c13	0.364	0.568	0.268	0.894	0.705	0.151†	ND	–
11	c09	0.267	0.555	0.212	0.695	0.693	11.8	ND	–
12	c09	0.261	0.641	0.271	0.685	-0.157	U	U	–
13	c13	0.389	0.538	0.236	0.614	0.616	0.769	56	72.8
14	c09	0.305	0.560	0.334	0.800	-0.096	1.79†	ND	–
15	c06	0.299	0.596	0.197	0.807	-0.154	0.254	ND	–
16	c06	0.808	0.336	0.223	0.411	0.411	7.03	ND	–
17	c09	0.719	0.560	0.371	0.702	0.703	2.17	ND	–
18	c09	0.692	0.542	0.253	0.701	0.701	U	U	–

Table 5 | **Platform-specific holo affinities and replicate variability.** Apparent K_D values are in nM, summarized by the arithmetic median and observed minimum–maximum across reported binding fits rated Good or Medium. Platforms are never pooled. A range describes replicate variability, not a confidence interval. $*$ marks an estimate extending beyond the tested concentration range. † marks a contributing rate-bound replicate. M13 is the positive control, measured by SPR only in this set. Its non-standard kinetics warrant treating its affinity as apparent.

Design	BLI n	BLI median [range]	SPR n	SPR median [range]
01	4	34 [31.7–38.1]	2	21.3 [20.7–21.9]
02	2	3.23 [3.06–3.4]	2	0.136 [0.135–0.138]
03	2	3.37 [3–3.75]	2	2.63 [2.57–2.68]
04	4	1,689 [617–3,529]*	2	439 [225–652]
05	2	2.04 [1.91–2.17]	2	7.63 [6.03–9.23]
06	4	8.25 [4.26–12.7]	2	2.52 [2.43–2.61]
07	0	ND	0	ND
08	2	4.41 [4.38–4.43]	2	6.5 [6.31–6.7]
09	4	3.53 [0.358–6.62]	2	0.539 [0.502–0.576]
10	1	0.151†	2	0.319 [0.229–0.409]
11	4	11.8 [4.05–25.2]	2	5.15 [5.07–5.23]
12	0	U	0	U
13	2	0.769 [0.71–0.829]	2	1.86 [1.85–1.87]
14	4	1.79 [0.175–3.43]†	2	0.699 [0.52–0.877]
15	2	0.254 [0.252–0.257]	2	0.1 [0.1–0.101]
16	4	7.03 [4.47–10.8]	2	4.38 [3.73–5.03]
17	4	2.17 [1.69–2.54]	2	2.3 [1.71–2.88]
18	0	U	0	U
M13	0	–	2	3.65 [3.4–3.9]

Table 6 | **Supplied conservative BLI-only interval estimates.** Central estimates and lower–upper limits are reproduced from the supplied interval analysis in nM. These are separate supplied summaries, not confidence intervals calculated for the medians in Table 5. Their nominal coverage and construction were not specified in the CSV. Mixed-platform intervals are not used. † marks the rate-bound estimate for #10, whose supplied interval is conditional on that bound. $*$ marks the extrapolated estimate for #04.

Design	Supplied BLI estimate	Supplied lower–upper limits
01	34.4	29.8–39.6
02	3.23	1.63–6.4
03	3.35	0.824–13.7
04	1,570*	503–4,920
05	2.04	0.92–4.51
06	7.39	2.93–18.6
08	4.41	4.07–4.77
09	1.77	0.157–20
10	0.151†	0.001–0.152
11	9.9	2.33–42.1
13	0.767	0.286–2.06
14	0.939	0.0872–10.1
15	0.254	0.224–0.29
16	6.74	3.16–14.4
17	2.11	1.55–2.86

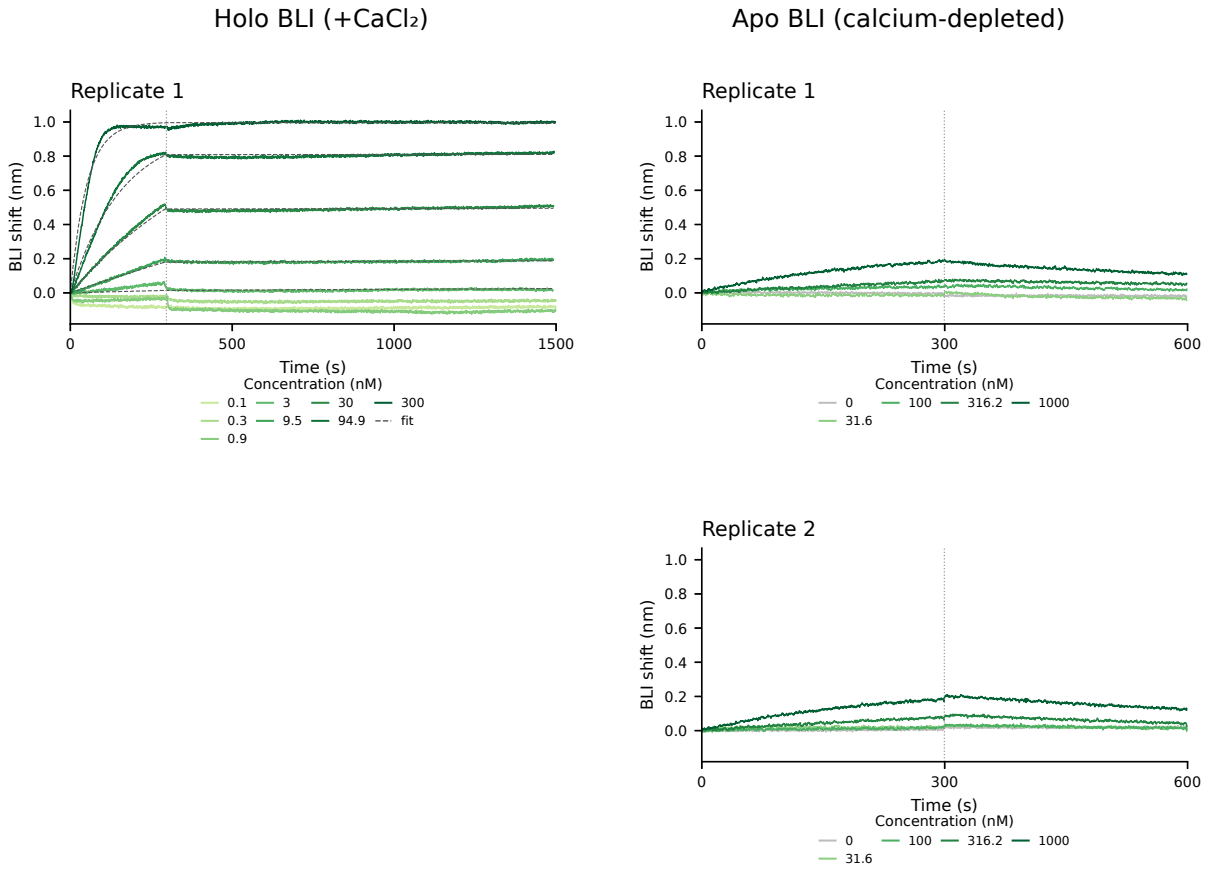


Figure 10 | **Individual BLI measurement series for design #10.** One holo BLI series is shown in the left column and two apo BLI series in the right column, with matched response scales. Main-figure panels D/E use series 1 in both conditions. The holo trace includes approximately 1,200 s of dissociation. Dashed curves show the selected fit, and dotted lines mark dissociation onset. Curves are supplied processed responses without additional fitting or smoothing.

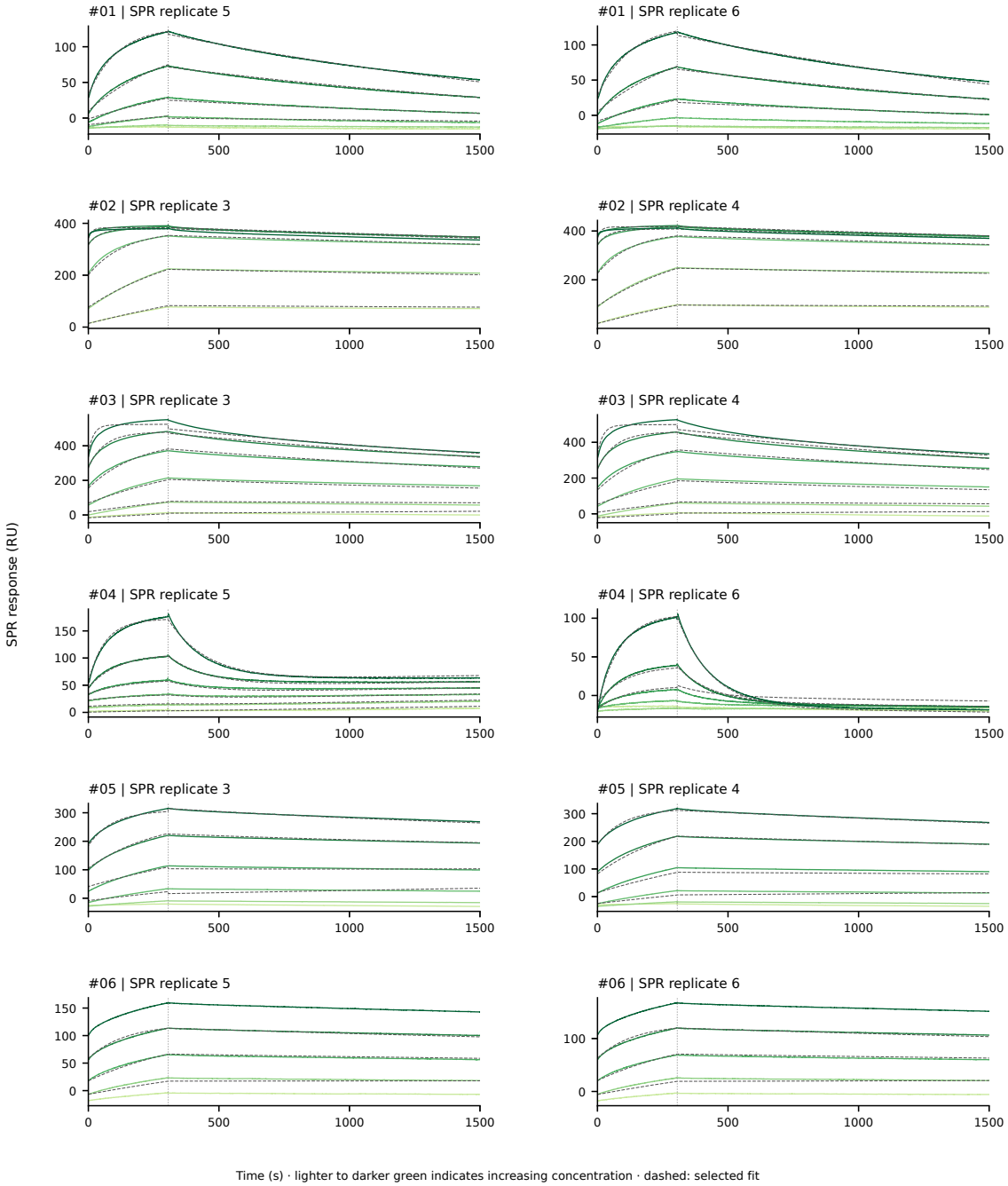


Figure 11 | **SPR measurements under calcium-loaded conditions.** The first two numbered SPR series are shown where available. Lighter to darker green denotes increasing concentration. Dashed curves show the supplied selected fits and dotted lines mark dissociation onset. SPR responses are in RU and are not pooled with BLI. The M13 control shows binding with non-standard kinetics.

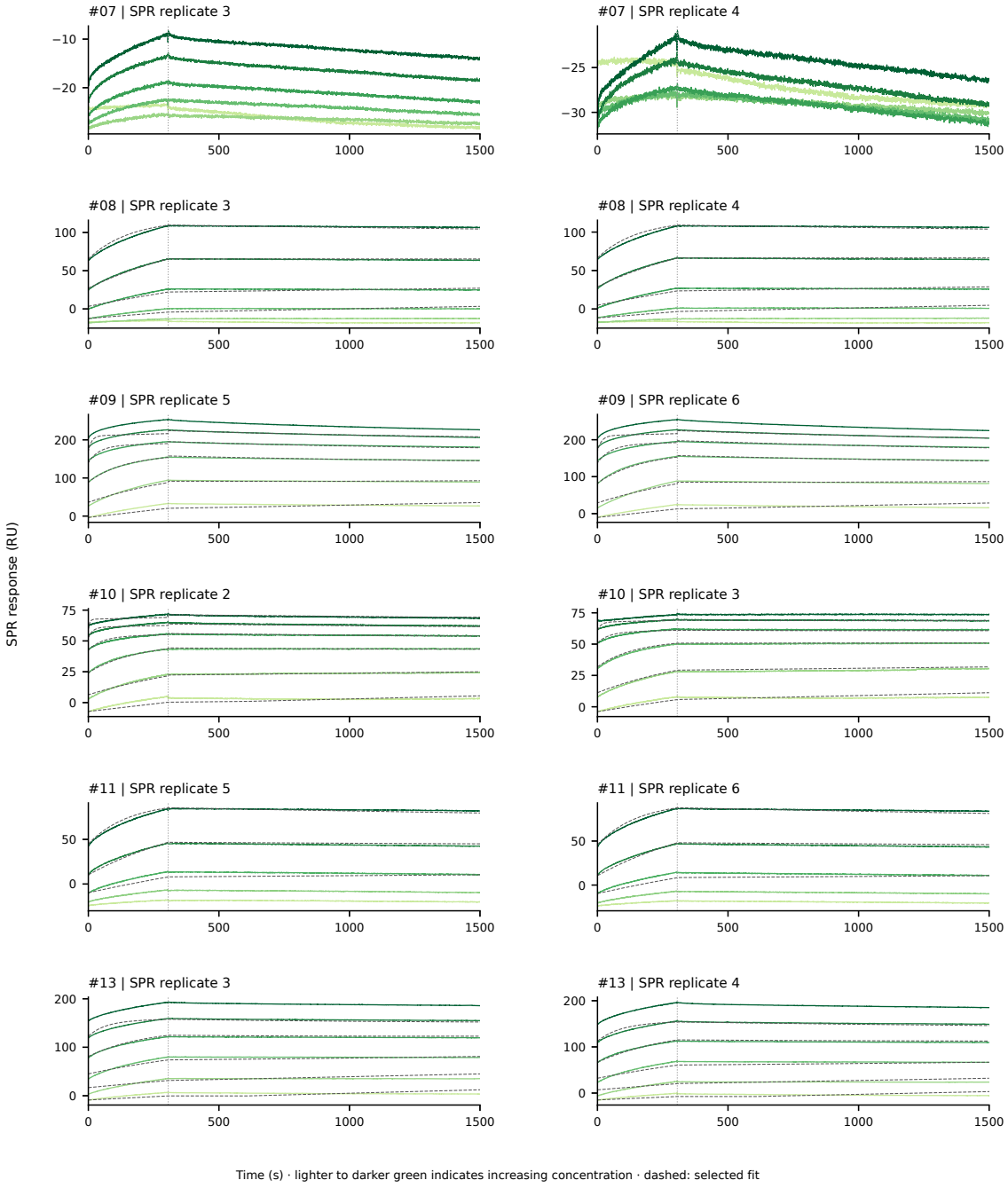


Figure 11 | **SPR measurements under calcium-loaded conditions (continued).** The first two numbered SPR series are shown where available. Lighter to darker green denotes increasing concentration. Dashed curves show the supplied selected fits and dotted lines mark dissociation onset. SPR responses are in RU and are not pooled with BLI. The M13 control shows binding with non-standard kinetics.

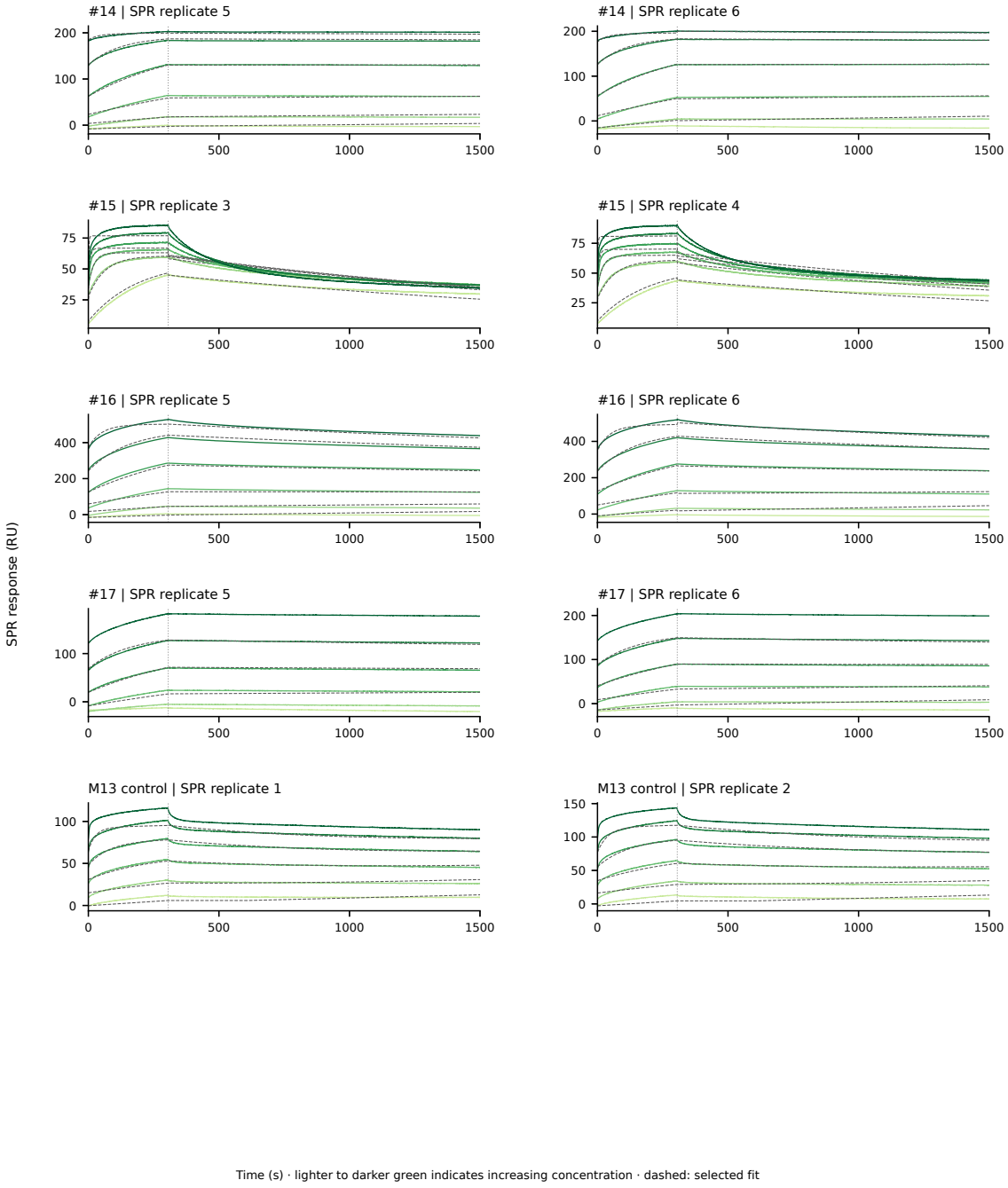


Figure 11 | **SPR measurements under calcium-loaded conditions (continued).** The first two numbered SPR series are shown where available. Lighter to darker green denotes increasing concentration. Dashed curves show the supplied selected fits and dotted lines mark dissociation onset. SPR responses are in RU and are not pooled with BLI. The M13 control shows binding with non-standard kinetics.

D. Assay definitions for the main examples

D.1. Paper-derived assay definitions and interpretation

Table 7 summarizes the design operations and computational comparisons for the five examples constructed with the assay builder in Fig. 4.

Table 7 | Design operations and computational comparisons for the five examples constructed with the assay builder in Fig. 4A–E.

Source / objective	Design operation	Computational comparison
Silva et al. (2019) IL-2/IL-15 mimic	De novo cytokine mimic	Dual-receptor engagement, receptor arrangement and helicity
Zhang et al. (2025) Hsp70/DnaK	De novo DnaK binder	Same candidate under ATP-/ADP-state conditioning
Huang et al. (2025) Endocytosis	Fusion of two de novo binders	Intact dual-domain engagement versus arm disruption
Dou et al. (2018) Fluorogen activation	mFAP1 scaffold redesign	DFHBI pose and pocket geometry versus pocket-Ala variants
Tinberg et al. (2013) Digoxigenin recognition	De novo small-molecule binder	Empirical-score preference over digitoxigenin, progesterone and β-estradiol

E. Autoresearch assessment criteria

The criteria below apply to the separate campaigns illustrated in Figure 5, not the experimental CaM cohort.

Table 8 | Fixed assessment criteria for the illustrated campaigns.

Campaign	Required measurements and criteria
Shared binding checks	For CaM, MCL-1, BCL-xL and DFHBI: designed-chain pTM $\geq 80/100$, minimum interface PAE ≤ 1.5 Å and binder backbone RMSD ≤ 2.5 Å. RMSD uses target alignment for protein targets and ligand alignment for DFHBI. All required comparisons must be valid.
CaM	Holo-minus-apo ipTM ≥ 0.1, desired-state ipTM ≥ 0.7 and positive AlloGen contrast (Cao et al., 2026). State-retention diagnostics remain advisory.
MCL-1/BIM	Target–binder ipTM ≥ 0.5, binder-minus-BIM ipTM > 0, native BIM-site overlap ≥ 0.25 against the MCL-1/BIM structure (Fire et al., 2010), and a valid competition comparison.
BCL-xL/KLHL20	Target and E3 ipTM ≥ 0.5 each, footprint overlap ≤ 0.10, no detected clash, valid internal-graft geometry and PDLV-motif contact with E3 (Mylemans et al., 2026). Footprint overlap is the fraction of fusion residues contacting either partner that contact both, using a 5 Å interatomic cutoff. Each revision used a fresh exact-construct OpenFold3 binary reference, Protenix-v2 binary assessment and OpenFold3 ternary assessment (Zhang et al., 2026; The OpenFold3 Team, 2026). Natural partners used verified homolog MSAs. The designed fusion used single-sequence inputs.
SAM/SAH	Valid chemistry, SAM mean-directed-minimum interface PAE ≤ 1.5 Å, pair ipTM ≥ 0.80, binder backbone RMSD ≤ 1.5 Å, protein-aligned ligand RMSD ≤ 5 Å, and SAM-favorable PAE and ipTM contrasts. Also required chain pTM $\geq 80/100$, audited minimum directed interface PAE ≤ 1.5 Å and ligand-aligned backbone RMSD ≤ 2.5 Å. Invalid or incomplete comparisons could not pass.
DFHBI (de novo)	Shared binding checks above.
DFHBI (mFAP1 redesign)	Planar Z conformation of DFHBI (Dou et al., 2018). Assay limits: conjugated-atom plane RMS deviation ≤ 0.1011 Å, phenyl–imidazolinone ring-plane angle $\leq 9.7359°$, and absolute Z dihedral $\leq 30°$.